Data Analysis

Lecture notes with R, Lectures 2–11

Contents

Lecture 2. Data Visualization 5

2.1 Why visualise data 5
2.1.1 Example: campaign spending in 2012 . . . 6

2.2 A taxonomy for data graphics 6

2.3 A directory of visualisations 7

2.4 Visualising distributions 7
2.4.1 Histograms and bin width . . . 7
2.4.2 Density estimates . . . 8
2.4.3 Comparing distributions . . . 9

2.5 Visualising associations 9

2.6 Visualising trends 10

2.7 Telling a story 11

Lecture 3. Data Wrangling 12

3.1 A grammar of data wrangling 12
3.1.1 The single-table verbs . . . 12

3.2 Relational data 14

3.3 Example: Uber pickups in New York 15

3.4 Selected exercises 15

Lecture 4. Statistical Foundations 17

4.1 From description to statistical methods 17

4.2 Samples and populations: a travel policy 17

4.3 Sampling distributions 18

4.4 The bootstrap 19

4.5 Outliers 20

4.6 Statistical models 21

4.7 Selected exercises 22

Lecture 5. Linear Regression **24**

5.1 Simple linear regression **24**
5.1.1 Example: the rail trail . . . 24
5.1.2 Prediction and intervals . . . 25
5.1.3 Strength of fit and least squares . . . 26
5.1.4 A categorical predictor . . . 26

5.2 Loss functions **26**

5.3 Multiple regression **27**
5.3.1 Inference for coefficients . . . 28
5.3.2 Choosing a model . . . 28

5.4 Other considerations **29**

5.5 Selected exercises **30**

Lecture 6. Generalized Linear Models and Model Evaluation **32**

6.1 Generalized linear models **32**

6.2 Logistic regression **32**
6.2.1 Coefficients as odds ratios . . . 33

6.3 GLMs for counts **33**
6.3.1 Poisson regression . . . 33
6.3.2 Overdispersion and the negative binomial . . . 34
6.3.3 Zero-inflated Poisson . . . 34

6.4 Evaluating models **35**
6.4.1 Training error and the holdout method . . . 35
6.4.2 Cross-validation . . . 36
6.4.3 Why a separate test set . . . 36

6.5 Selected exercises **36**

Lecture 7. Regularization **39**

7.1 Beyond least squares **39**

7.2 Subset selection **39**
7.2.1 Choosing the optimal model . . . 40

7.3 Regularization **40**
7.3.1 Penalized form . . . 41

7.4 Example: delays of flights to California **41**

7.5 Selected exercises **43**

Lecture 8. Classification I 46

8.1 Supervised learning 46

8.2 Naive Bayes 46

8.3 Discriminant analysis 47

8.4 Logistic regression for classification 48
8.4.1 Why not squared loss . . . 49
8.4.2 Cross-entropy loss . . . 50

8.5 Selected exercises 50

Lecture 9. Classification II 52

9.1 Logistic regression as a classifier 52
9.1.1 Thresholding . . . 52
9.1.2 Accuracy . . . 52

9.2 Evaluating classifiers 52
9.2.1 Drawbacks of accuracy . . . 52
9.2.2 The confusion matrix . . . 53
9.2.3 ROC and precision–recall curves . . . 53
9.2.4 Training versus testing . . . 54

9.3 Linear separability and regularization 55

9.4 Selected exercises 56

Lecture 10. Splines and Generalized Additive Models 58

10.1 Beyond linearity 58

10.2 Polynomial regression and step functions 58

10.3 Regression splines 59

10.4 Smoothing splines 60

10.5 Local regression 61

10.6 Generalized additive models 62

10.7 Selected exercises 63

Lecture 11. Causal Inference 65

11.1 Why causation 65

11.2 Potential outcomes 65

11.3 Randomised experiments and observational studies 66
11.3.1 Unconfoundedness . . . 66

11.4 Propensity score methods **67**
11.4.1 Stratification 67
11.4.2 Matching 67
11.4.3 Inverse probability weighting 67

11.5 Example: job training and earnings **68**

11.6 Selected exercises **69**

Lecture 2. Data Visualization

These notes follow the course slides (based on *Modern Data Science with R*, Baumer, Kaplan and Horton, and Wilke's *Fundamentals of Data Visualization*). Lines starting with ## are R output; data sets come from R packages. Material marked [Supplement] is not on the slides.

2.1 Why visualise data

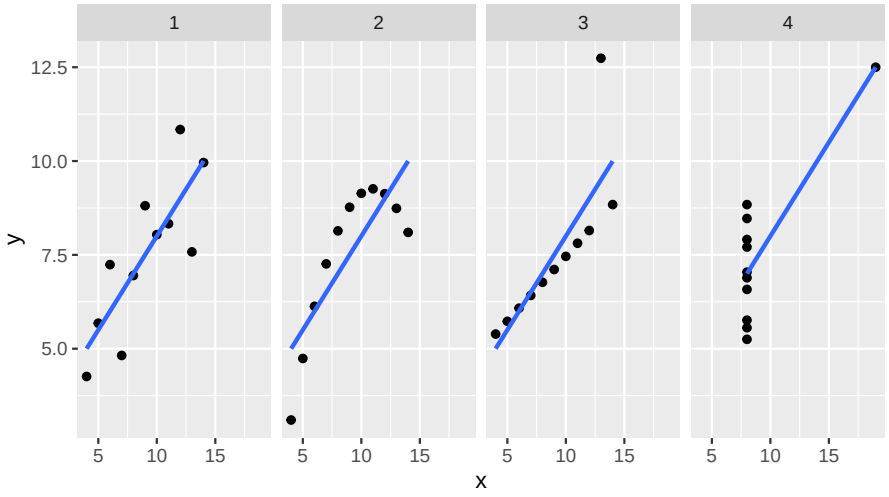
Motivation. Very different data sets can share the same summary statistics; a plot shows what the numbers hide.

Anscombe's quartet: four data sets with (almost) identical means, variances, correlations and regression lines.

```
anscombe_long <- anscombe %>%
  pivot_longer(everything(), names_to = c(".value", "set"), names_pattern = "(.)(.)")
anscombe_long %>%
  group_by(set) %>%
  summarise(mean_x = mean(x), mean_y = mean(y), sd_y = sd(y), cor = cor(x, y))
```

```
## # A tibble: 4 x 5
##   set   mean_x mean_y  sd_y   cor
##   <chr>  <dbl>  <dbl> <dbl> <dbl>
## 1 1          9   7.50  2.03 0.816
## 2 2          9   7.50  2.03 0.816
## 3 3          9   7.5   2.03 0.816
## 4 4          9   7.50  2.03 0.817
```

```
ggplot(anscombe_long, aes(x, y)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, formula = y ~ x) +
  facet_wrap(~ set, nrow = 1)
```



Only set 1 is linear; set 2 is a curve, set 3 has one outlier, set 4 is driven by a single point. Visualisation "forces us to notice what we never expected to see" (Tukey, 1977).

There is no fixed procedure: look from several perspectives and levels of detail, start simply, refine. Tufte's principles: above all show the data; avoid distorting what the data have to say; present many numbers in a small space; encourage the eye to compare; reveal the data at several levels of detail.

Main tools: histograms, densities, bar plots, boxplots (distributions); scatterplots (bivariate data); heat maps and maps (specialised).

2.1.1 Example: campaign spending in 2012

US Federal Election Commission spending data, 2012 (slides). Three lessons:

- **A third variable can reverse the story.** Total spending made Obama look best funded, but colouring by type of spending showed that $251 million of the $274 million spent on his campaign was spent *against* him (92%, against 70% for Romney).
- **Faceting adds context.** By office (House, Senate, President): only in the presidential race was more spent attacking Democrats than Republicans.
- **Proportions, not totals.** In House races compare vote share with the *share* of spending (districts differ in turnout). Use a log scale for skewed amounts; overlay or stack distributions to compare them.

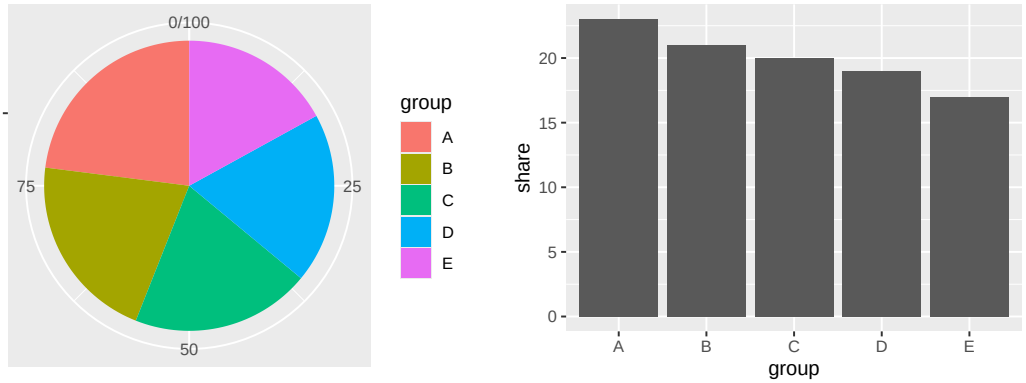
2.2 A taxonomy for data graphics

Motivation. Yau's four elements (Yau, 2013), matching ggplot2's grammar of graphics, break any graphic into parts.

1. **Visual cues**: graphical elements that draw the eye. In decreasing order of how accurately people perceive differences: position, length, angle, direction, shape (categorical), area, volume, shade and colour.
2. **Coordinate system**: Cartesian, polar, or geographic.
3. **Scale**: how distance in the graphic translates into differences in quantity. Numeric scales may be linear, logarithmic or percentage; categorical scales may be unordered or ordinal; time is numeric but has calendar units and can be periodic.
4. **Context**: titles, axis labels, reference points or lines that help the viewer make meaningful comparisons.

Positions and lengths are judged well, angles and areas poorly: **bar charts beat pie charts**; bubble sizes and colour differences convey magnitudes only roughly; red–green fails for colour-blind viewers.

```
shares <- tibble(group = LETTERS[1:5], share = c(23, 21, 20, 19, 17))
ggplot(shares, aes(x = "", y = share, fill = group)) +
  geom_col(width = 1) + coord_polar(theta = "y") + labs(x = NULL, y = NULL)
ggplot(shares, aes(group, share)) + geom_col()
```



Same five shares: equal-looking slices, obvious bar ordering.

Dissecting a graphic. Ask: which variables (and types), which cue for each, which coordinates and scales, what context. E.g. mean SAT score by state: continuous score and categorical state, length cue, Cartesian, linear, axis labels and title.

2.3 A directory of visualisations

Motivation. Most plotting questions fall into a few types, each with a standard plot.

what to show	plots
amounts	bars (vertical or horizontal), dots where the bar would end
one distribution	histogram, density (need a binwidth or bandwidth); empirical CDF, QQ plot (faithful but harder to read)
several distributions	boxplots, violins, overlapping densities, ridgeline plots (many groups)
proportions	side-by-side bars, stacked bars (many conditions), pies (only for simple fractions of a whole); stacked densities when proportions change along a continuous variable
multiple groupings	mosaic plots, treemaps, parallel sets
x–y relationships	scatterplots, bubble charts (third variable as size), 2-d bins for many points, correlograms for many variables, line graphs when x is time
geospatial	choropleth maps, cartograms

2.4 Visualising distributions

Motivation. Histograms and densities depend on a bin width or bandwidth; standard rules give a starting point.

2.4.1 Histograms and bin width

Bin width h: too narrow shows noise, too wide hides structure (slides: Titanic ages, widths 1, 3, 5, 15). Two rules:

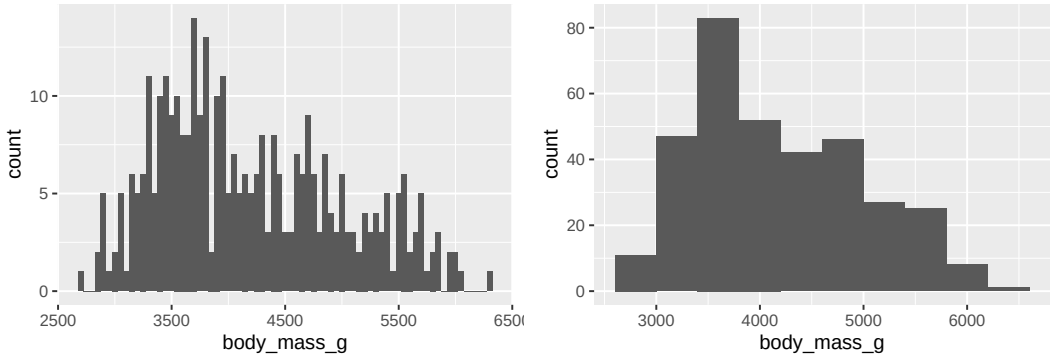
$$\text{Scott: } h = 3.49\,\hat{\sigma}\,n^{-1/3}, \qquad \text{Freedman–Diaconis: } h = 2\,\text{IQR}\,n^{-1/3},$$

the second robust to outliers. Penguin body mass:

```
x <- na.omit(penguins$body_mass_g)
c(n = length(x), scott = 3.49 * sd(x) * length(x)^(-1/3),
  fd = 2 * IQR(x) * length(x)^(-1/3))
```

```
##        n    scott       fd
## 342.0000 400.2209 343.1910
```

```
ggplot(penguins, aes(body_mass_g)) + geom_histogram(binwidth = 50, na.rm = TRUE)
ggplot(penguins, aes(body_mass_g)) + geom_histogram(binwidth = 400, na.rm = TRUE)
```



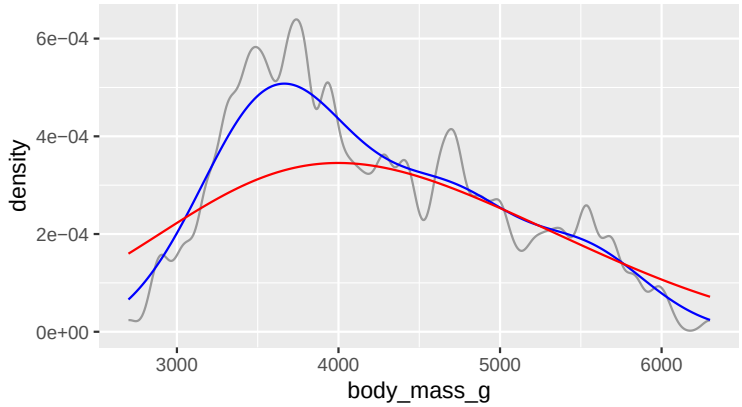
2.4.2 Density estimates

A kernel density estimate sums a bump (**kernel** K) at each observation:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big).$$

The bandwidth h matters far more than K. Rule of thumb (Härdle, 1991): $h = 1.06\, S\, n^{-1/5}$, where S is the standard deviation; robust versions replace S by the scaled IQR $(X_{(0.75n)} - X_{(0.25n)})/1.34$ or the scaled MAD $1.4826 \cdot \text{median}|X_i - \text{median}(X)|$.

```
ggplot(penguins, aes(body_mass_g)) +
  geom_density(bw = 50, colour = "grey60", na.rm = TRUE) +
  geom_density(bw = 1.06 * sd(x) * length(x)^(-1/5), colour = "blue", na.rm = TRUE) +
  geom_density(bw = 800, colour = "red", na.rm = TRUE)
```



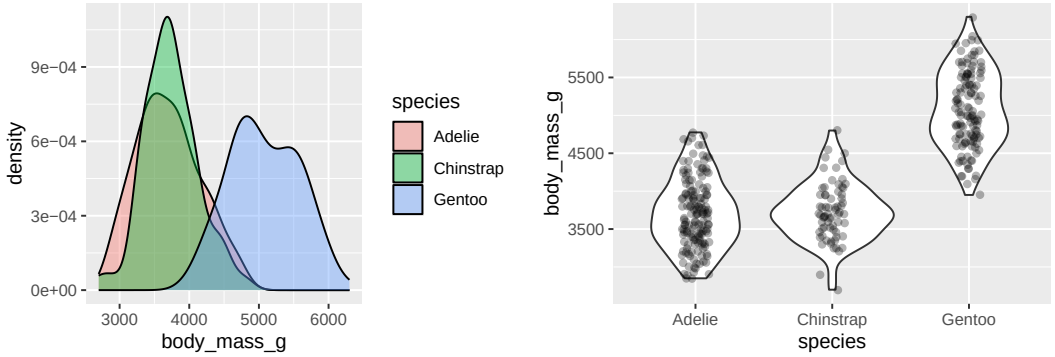
Where the 1.06 comes from [Supplement]

For a normal density with standard deviation σ and a Gaussian kernel, the bandwidth minimising the asymptotic mean integrated squared error is $h^* = (4/3)^{1/5}\sigma n^{-1/5} \approx 1.06\,\sigma n^{-1/5}$. It oversmooths multimodal data (the blue curve blurs the species).

2.4.3 Comparing distributions

Two distributions: overlay (with transparency). Several: jittered points, boxplots, violins, or ridgelines (`ggridges`, many groups).

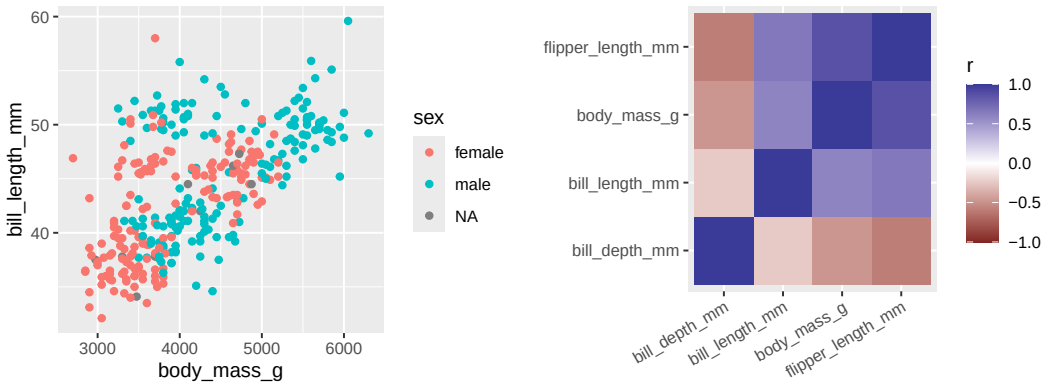
```
ggplot(penguins, aes(body_mass_g, fill = species)) +
  geom_density(alpha = 0.4, na.rm = TRUE)
ggplot(penguins, aes(species, body_mass_g)) +
  geom_violin(na.rm = TRUE) + geom_jitter(width = 0.1, alpha = 0.3, na.rm = TRUE)
```



2.5 Visualising associations

(Slides: 123 blue jays.) Scatterplot for two variables; colour for a group; size for a third variable (**bubble chart**, hard to judge); scatterplot matrix; **correlogram** (picture of the correlation matrix).

```
ggplot(penguins, aes(body_mass_g, bill_length_mm, colour = sex)) +
  geom_point(na.rm = TRUE)
penguins %>%
  select(bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g) %>%
  cor(use = "complete.obs") %>%
  as_tibble(rownames = "x") %>%
  pivot_longer(-x, names_to = "y", values_to = "r") %>%
  ggplot(aes(x, y, fill = r)) + geom_tile() +
  scale_fill_gradient2(limits = c(-1, 1)) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1)) + labs(x = NULL, y = NULL)
```



`GGally::ggpairs()` draws the full scatterplot matrix in one call.

Simpson's paradox in the penguins [Supplement]

Pooled over species, bill depth and bill length are negatively correlated; within each species the correlation is positive. A grouping variable can reverse an association.

```
penguins %>% drop_na() %>%
  summarise(pooled = cor(bill_length_mm, bill_depth_mm))
```

```
## # A tibble: 1 x 1
##   pooled
##    <dbl>
## 1 -0.229
```

```
penguins %>% drop_na() %>% group_by(species) %>%
  summarise(within = cor(bill_length_mm, bill_depth_mm))
```

```
## # A tibble: 3 x 2
##   species   within
##   <fct>      <dbl>
## 1 Adelie     0.386
## 2 Chinstrap  0.654
## 3 Gentoo     0.654
```

2.6 Visualising trends

Motivation. Smoothing separates trend from noise; a transformation can straighten a curved trend.

Defined functional forms. bioRxiv submissions grow as $y = Ae^{mx}$; fit a line to $\log y = \log A + mx$.

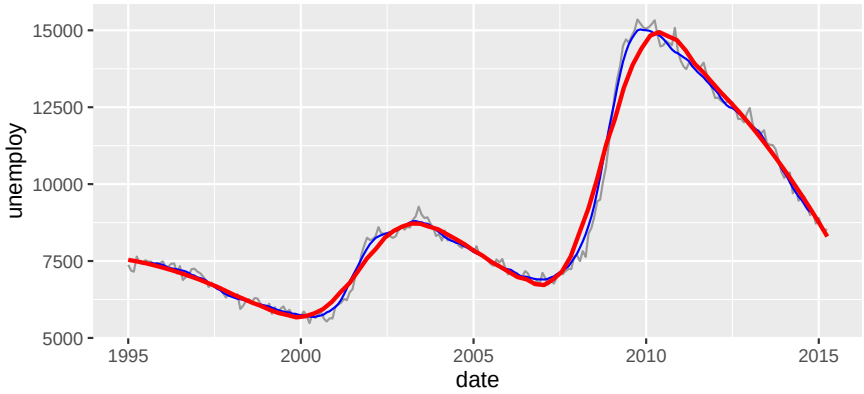
Smoothing. (Slides: Dow Jones 2009.)

- a **moving average**: the mean over a window of neighbouring times;
- **LOESS** (locally estimated scatterplot smoothing): at each x_0, take the k nearest observations $N_k(x_0)$ and fit a line $\ell(x) = \beta_0 + \beta_1(x - x_0)$ by weighted least squares, minimising

$$\sum_{i:X_i \in N_k(x_0)} \{Y_i - \ell(X_i)\}^2 \, W\Big(\frac{|x_0 - X_i|}{\Delta(x_0)}\Big),$$

  where $\Delta(x_0) = \max_{i:X_i \in N_k(x_0)} |X_i - x_0|$ and $W(u) = (1 - u^3)^3$ (tri-cube weight). The **span** k/n controls smoothness; a quadratic can replace the line;
- **splines**: piecewise polynomials joined smoothly at *knots*; faster than LOESS, but sensitive to type and number of knots (Lecture 10).

```
econ <- economics %>%
  filter(date >= as.Date("1995-01-01")) %>%
  mutate(ma12 = stats::filter(unemploy, rep(1/12, 12), sides = 2))
ggplot(econ, aes(date, unemploy)) +
  geom_line(colour = "grey60") +
  geom_line(aes(y = ma12), colour = "blue", na.rm = TRUE) +
  geom_smooth(method = "loess", span = 0.3, se = FALSE, colour = "red", formula = y ~ x)
```



US unemployment (thousands, `ggplot2::economics`): raw series (grey), 12-month moving average (blue) and LOESS with span 0.3 (red).

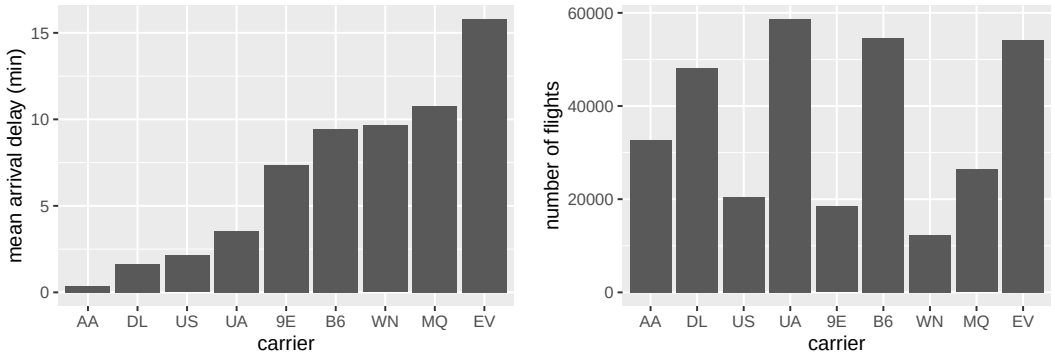
2.7 Telling a story

Motivation. The Challenger disaster is the classic case of the right data shown the wrong way.

On 27 January 1986 Morton Thiokol engineers urged delaying the Challenger launch, fearing cold would compromise the O-rings; their 13 charts did not persuade NASA, and the O-rings failed 73 seconds after launch. The charts never plotted damage against temperature, and the hearings showed only damaged flights. Tufte's plot of damage against temperature for *all* flights: every launch below about 65°F had damage, and the forecast (about 30°F) was far below every previous launch.

Making a point simply. For the message "American and Delta have the shortest delays", a bar chart beats a cluttered delay-over-time plot; flight counts show both are major NYC carriers:

```
by_carrier <- flights %>%
  group_by(carrier) %>%
  summarise(delay = mean(arr_delay, na.rm = TRUE), n = n()) %>%
  filter(n > 10000)
ggplot(by_carrier, aes(reorder(carrier, delay), delay)) + geom_col() +
  labs(x = "carrier", y = "mean arrival delay (min)")
ggplot(by_carrier, aes(reorder(carrier, delay), n)) + geom_col() +
  labs(x = "carrier", y = "number of flights")
```



Lecture 3. Data Wrangling

3.1 A grammar of data wrangling

Motivation. dplyr gives a grammar for data: a few verbs, each taking and returning a data frame, chained together.

`dplyr` is fast on in-memory data and separates operations from the data source (the same code runs on a database). It works on data frames and **tibbles** (class `tbl_df`), which print only the first rows and fitting columns, with types. Vignette: `vignette("dplyr", package = "dplyr")`.

We use `ggplot2::presidential`:

```
presidential
```

```
## # A tibble: 12 x 4
##   name       start      end        party
##   <chr>      <date>     <date>     <chr>
## 1 Eisenhower 1953-01-20 1961-01-20 Republican
## 2 Kennedy    1961-01-20 1963-11-22 Democratic
## 3 Johnson    1963-11-22 1969-01-20 Democratic
## 4 Nixon      1969-01-20 1974-08-09 Republican
## 5 Ford       1974-08-09 1977-01-20 Republican
## # i 7 more rows
```

```
class(presidential)
```

```
## [1] "tbl_df"     "tbl"        "data.frame"
```

3.1.1 The single-table verbs

- `filter()` selects rows; `select()` selects columns;
- `mutate()` adds columns; `arrange()` reorders rows;
- `summarize()` collapses a data frame to a single row, usually together with `group_by()`.

filter() keeps the rows where a logical vector is `TRUE`. Comparisons: `>`, `>=`, `<`, `<=`, `!=`, `==`; set membership `x %in% Y`; and they combine with `&`, `|`, `!`.

```
x <- c(2, 1, 3, 0)
x == 0 | x == 1
```

```
## [1] FALSE  TRUE FALSE  TRUE
```

```
x %in% c(0, 1)
```

```
## [1] FALSE  TRUE FALSE  TRUE
```

```
filter(presidential, party == "Republican")
```

```
## # A tibble: 7 x 4
##   name       start      end        party
##   <chr>      <date>     <date>     <chr>
## 1 Eisenhower 1953-01-20 1961-01-20 Republican
## 2 Nixon      1969-01-20 1974-08-09 Republican
## 3 Ford       1974-08-09 1977-01-20 Republican
## 4 Reagan     1981-01-20 1989-01-20 Republican
## 5 Bush       1989-01-20 1993-01-20 Republican
## # i 2 more rows
```

Missing values are contagious: `NA > 5`, `NA + 10`, `NA == NA` are `NA`. Test with `is.na()`; drop with `filter(!is.na(variable))`.

select() picks columns; helpers `starts_with()`, `ends_with()`, `contains()`, `num_range()` select by name pattern:

```
flights %>% select(ends_with("time"))
```

```
## # A tibble: 336,776 x 5
##   dep_time sched_dep_time arr_time sched_arr_time air_time
##      <int>          <int>    <int>          <int>    <dbl>
## 1      517            515      830            819      227
## 2      533            529      850            830      227
## 3      542            540      923            850      160
## 4      544            545     1004           1022      183
## 5      554            600      812            837      116
## # i 336,771 more rows
```

The pipe. Which Democratic presidents served since Watergate? Nesting `select(filter(presidential, start > 1973 & party == "Democratic"), name)` reads inside out. The pipe `%>%` (Ctrl+Shift+M) passes the left side as first argument: `df %>% filter(cond)` = `filter(df, cond)`:

```
presidential %>%
  filter(start > 1973 & party == "Democratic") %>%
  select(name)
```

```
## # A tibble: 3 x 1
##   name
##   <chr>
## 1 Carter
## 2 Clinton
## 3 Obama
```

mutate() creates variables (term length in years via lubridate):

```
my_presidents <- presidential %>%
  mutate(term_length = interval(start, end) / dyears(1))
```

Save as a new object rather than overwrite external data.

arrange() sorts rows; later keys break ties:

```
my_presidents %>% arrange(desc(term_length), party)
```

```
## # A tibble: 12 x 5
##   name       start      end        party      term_length
##   <chr>      <date>     <date>     <chr>            <dbl>
## 1 Clinton    1993-01-20 2001-01-20 Democratic           8
## 2 Obama      2009-01-20 2017-01-20 Democratic           8
## 3 Eisenhower 1953-01-20 1961-01-20 Republican           8
## 4 Reagan     1981-01-20 1989-01-20 Republican           8
## 5 Bush       2001-01-20 2009-01-20 Republican           8
## # i 7 more rows
```

summarize() and group_by(). `summarize()` reduces columns to one value; `n()` counts rows, `sum(party == "Democratic")` counts TRUEs.

```
my_presidents %>%
  summarize(N = n(), first_year = min(year(start)), last_year = max(year(end)),
            num_dems = sum(party == "Democratic"),
            years = sum(term_length), avg_term_length = mean(term_length))
```

```
## # A tibble: 1 x 6
##       N first_year last_year num_dems years avg_term_length
##   <int>      <dbl>     <dbl>    <int> <dbl>           <dbl>
## 1    12       1953      2021        5    68            5.67
```

Units [Supplement]

The slides divide by 365.25 and get 0.186; `term_length` is already in years, so the total is 68 years (1953–2021).

Grouping runs the summary per party:

```
my_presidents %>%
  group_by(party) %>%
  summarize(N = n(), avg_term_length = mean(term_length),
            std_term_length = sd(term_length))
```

```
## # A tibble: 2 x 4
##   party          N avg_term_length std_term_length
##   <chr>      <int>           <dbl>           <dbl>
## 1 Democratic     5            5.6             2.34
## 2 Republican     7            5.71            2.32
```

Average terms hardly differ between the parties.

3.2 Relational data

Motivation. Databases keep each fact in one table, linked by keys; analyses join them back.

Relational data: linked tables. Two-table verbs: *mutating joins* (add variables from matching rows), *filtering joins* (keep/drop rows by match), *set operations*. In `nycflights13`, `flights` stores carrier codes and `airlines` maps them to names. Keys: `flights` connects to `airlines` by `carrier`, to `planes` by `tailnum`, to `airports` by `origin` and `dest`, and to `weather` by `origin` and `year`, `month`, `day`, `hour`.

inner_join() keeps rows with a match in both tables:

```
flights_joined <- flights %>%
  inner_join(airlines, by = c("carrier" = "carrier"))
flights_joined %>% select(carrier, name, flight, origin, dest)
```

```
## # A tibble: 336,776 x 5
##   carrier name                  flight origin dest
##   <chr>   <chr>                  <int> <chr>  <chr>
## 1 UA      United Air Lines Inc.   1545 EWR    IAH
## 2 UA      United Air Lines Inc.   1714 LGA    IAH
## 3 AA      American Airlines Inc.  1141 JFK    MIA
## 4 B6      JetBlue Airways          725 JFK    BQN
## 5 DL      Delta Air Lines Inc.     461 LGA    ATL
## # i 336,771 more rows
```

left_join() keeps all rows of the first table, `NA` where unmatched. Pacific-time airports:

```
airports_pt <- airports %>% filter(tz == -8)
flights %>% inner_join(airports_pt, by = c("dest" = "faa")) %>% nrow()
```

```
## [1] 46324
```

```
flights %>%
  left_join(airports_pt, by = c("dest" = "faa")) %>%
  summarize(num_flights = n(),
            num_flights_pt = sum(!is.na(name)),
            num_flights_not_pt = sum(is.na(name)))
```

```
## # A tibble: 1 x 3
##   num_flights num_flights_pt num_flights_not_pt
##         <int>          <int>              <int>
## 1      336776          46324             290452
```

Inner join: 46 324 west-coast flights; left join: all flights, others with `NA` name. Other joins: `right_join()` keeps all rows of the second table, `full_join()` all rows of both, `semi_join()` the rows of the first table that have a match (without adding columns), and `anti_join()` those that do not.

3.3 Example: Uber pickups in New York

Motivation. Combine, parse, derive, group, summarise, plot.

About 4.5 million NYC Uber pickups, April–September 2014: six CSV files with `Date/Time`, `Lat`, `Lon`, `Base` (company code). Code shown without output.

```
files <- sprintf("data/uber-raw-data-%s14.csv", c("apr", "may", "jun", "jul", "aug", "
    sep"))
data_2014 <- map_dfr(files, read_csv) %>%          # stack the six months
  mutate(dt = mdy_hms(`Date/Time`),
         month = month(dt, label = TRUE), day = day(dt),
         dayofweek = wday(dt, label = TRUE), hour = hour(dt))
```

(Slides: `rbind()`, `as.POSIXct()`; `map_dfr()` and lubridate are shorter [Supplement].) Each question is a grouped count plus bar chart:

```
data_2014 %>%
  count(hour, name = "Total") %>%
  ggplot(aes(hour, Total)) +
  geom_col(fill = "steelblue") +
  scale_y_continuous(labels = scales::comma) +
  ggtitle("Trips every hour")
data_2014 %>%
  count(month, dayofweek, name = "Total") %>%
  ggplot(aes(month, Total, fill = dayofweek)) +
  geom_col(position = "dodge")
```

Trips by hour, month, day and weekday follow this pattern. `Lon` against `Lat` (latitude 40.58–40.92, longitude −74.15 to −73.70) draws a map of New York; colour by `Base` for company areas, labelled via `left_join(by = c("Base" = "base_code"))`.

Overplotting millions of points [Supplement]

For 4.5 million points use `size = 0.1, alpha = 0.05` or 2-d bins (`geom_bin2d()`, `geom_hex()`).

3.4 Selected exercises

Motivation. Join sizes, missing-value logic and grouped means, proved once.

Exercise 3.1 (size of a join)

Let `x` and `y` be tables joined by a key. For a key value k let a_k and b_k be the numbers of rows of `x` and of `y` with key k (by default dplyr treats a missing key as one more key value, so `NA` matches `NA`).
(a) Show that `inner_join(x, y)` has $\sum_k a_k b_k$ rows and `left_join(x, y)` has $\sum_k a_k \max(b_k, 1)$ rows.

(b) Deduce that `left_join(x, y)` has exactly as many rows as `x` if and only if every key value of `x` occurs at most once in `y`.
(c) Show that `semi_join(x, y)` and `anti_join(x, y)` split the rows of `x` into two disjoint parts, so their row counts add up to that of `x`.

Solution [Supplement].

(a) An inner join outputs one row per pair (row of `x`, row of `y`) with equal keys: $a_k b_k$ pairs for key k, disjoint across k, total $\sum_k a_k b_k$. A left join adds one `NA`-filled row per unmatched row of `x`: for key k, $a_k b_k$ rows if $b_k \geq 1$ and a_k if $b_k = 0$, i.e. $a_k \max(b_k, 1)$.

(b) By (a) the difference from $\sum_k a_k$ is $\sum_k a_k\{\max(b_k, 1) - 1\}$, a sum of terms ≥ 0; it is 0 iff $b_k \leq 1$ for every k with $a_k > 0$.

(c) A semi join keeps a row of `x` once if $b_k \geq 1$, an anti join once if $b_k = 0$; exactly one holds, so each row appears once in exactly one result (a semi join never duplicates rows, unlike an inner join with $b_k \geq 2$). ■

Exercise 3.2 (logic of missing values; adapted from R4DS, Exercise 5.2.4.4)

`NA` stands for a value that exists but is unknown. Explain why `NA^0` is 1, `NA | TRUE` is `TRUE` and `FALSE & NA` is `FALSE`, while `NA > 5` and `NA * 0` are `NA`. State the general rule.

Solution [Supplement].

Step 1. (Rule.) R returns a known value iff the result is the same for every value the unknown could take.

Step 2. (Known results.) For every real u (including $\pm\infty$ and `NaN` in R's arithmetic) $u^0 = 1$, so `NA^0` is 1. For a logical u, $u \vee \text{TRUE} = \text{TRUE}$ and $\text{FALSE} \wedge u = \text{FALSE}$ whatever u is.

Step 3. (Unknown results.) `NA > 5` is `TRUE` for $u = 6$, `FALSE` for $u = 4$. For `NA * 0`, $u \cdot 0 = 0$ for finite u but $\infty \cdot 0$ is `NaN`, so the result is undetermined: the rule covers all representable values, not only finite ones. ■

Exercise 3.3 (combining grouped means)

Split n rows into groups $g = 1, \ldots, G$ of sizes n_g with group means $\bar{y}_g$ of a variable y.
(a) Show that the overall mean is $\bar{y} = \sum_g (n_g/n)\, \bar{y}_g$, so that the unweighted mean of the group means equals $\bar{y}$ when all n_g are equal or all $\bar{y}_g$ are equal, but not in general.
(b) Two groups A and B are each split by a binary variable s. In A, $s = 1$ for 80 of 100 rows with mean 10 and $s = 0$ for 20 rows with mean 2; in B, $s = 1$ for 20 of 100 rows with mean 11 and $s = 0$ for 80 rows with mean 3. Compare A and B within each level of s and overall.

Solution [Supplement].

(a) $\sum_i y_i = \sum_g \sum_{i \in g} y_i = \sum_g n_g \bar{y}_g$; divide by $n = \sum_g n_g$. Then

$$\bar{y} - \frac{1}{G} \sum_g \bar{y}_g = \sum_g \Big(\frac{n_g}{n} - \frac{1}{G} \Big) \bar{y}_g,$$

which is 0 for equal sizes or for $\bar{y}_g \equiv c$ (then $c \sum_g (n_g/n - 1/G) = 0$), but not in general.

(b) Within levels B is higher ($11 > 10$, $3 > 2$). Overall, by (a), A: $0.8 \cdot 10 + 0.2 \cdot 2 = 8.4$, B: $0.2 \cdot 11 + 0.8 \cdot 3 = 4.6$, so A is higher. The reversal (Simpson's paradox) comes from the different shares of the high-mean level $s = 1$. ■

Lecture 4. Statistical Foundations

4.1 From description to statistical methods

Motivation. To go beyond the data in hand we must know how much the numbers would change with different data.

Statistical methods quantify patterns and their strength, find patterns too complex to see, and interpret data through models. **Exploratory** analysis checks integrity and generates hypotheses (transform–visualise–model); **confirmatory** analysis checks a model or hypothesis, for prediction or inference. Data used for exploration are not reused for confirmation.

The t-test for $H_0 : \mu = 0$ rejects when $|t| > 1.96$, $t = \sqrt{n}\,\bar{X}/s$; behind it lie

- a **model**: observations centred at an unknown μ with random error;
- a **fit**: centre and error estimated by $\bar{X}$ and s;
- **assumptions**: numeric, (approximately) normal observations;
- **uncertainty quantification**: the 95% confidence interval $\bar{X} \pm 1.96\, s/\sqrt{n}$;
- **strength of evidence**: the smaller the p-value, the stronger the evidence against the hypothesis.

Key idea: the data are a **sample** from a **population**; imagine the other samples that might have been drawn.

4.2 Samples and populations: a travel policy

Motivation. Knowing the whole population shows how much a small sample can mislead.

How early should a New York traveller's flight be scheduled to arrive in San Francisco so that delays rarely make them late? Treat the 2013 SFO flights as the population:

```
SF <- flights %>% filter(dest == "SFO", !is.na(arr_delay))
set.seed(101)
sf_25 <- SF %>% slice_sample(n = 25)
sf_25 %>% fivenum_tbl(arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1    25  0.12  29.8   -38   -23     -5    14   103
```

```
SF %>% fivenum_tbl(arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1 13173  2.67  47.7   -86   -23     -8    12  1007
```

(`set.seed()` makes `slice_sample()` reproducible.) The sample maximum is under two hours; the population has 1007 minutes. Accept a 2% risk, i.e. use the 98th percentile:

```
sf_25 %>% summarize(q98 = quantile(arr_delay, p = 0.98))
```

```
## # A tibble: 1 x 1
##     q98
##   <dbl>
## 1  67.5
```

```
SF %>% group_by(arr_delay < 90) %>% count() %>% mutate(pct = n / nrow(SF))
```

```
## # A tibble: 2 x 3
## # Groups:   arr_delay < 90 [2]
##   `arr_delay < 90`     n    pct
##   <lgl>            <int>  <dbl>
## 1 FALSE              640 0.0486
## 2 TRUE             12533 0.951
```

```
SF %>% summarize(q98 = quantile(arr_delay, p = 0.98))
```

```
## # A tibble: 1 x 1
##     q98
##   <dbl>
## 1   153
```

A 90-minute policy fails about 5% of the time; the population says about 150 minutes. The question is how good a sample estimate is.

4.3 Sampling distributions

Motivation. A statistic is random; its standard error measures its reliability.

A **statistic** summarises data (mean, sd, median); extremes such as the maximum are unreliable. Draw many samples from the population:

```
n <- 25
num_trials <- 500
sf_25_means <- 1:num_trials %>%
  map_dfr(~ SF %>% slice_sample(n = n) %>% summarize(mean_arr_delay = mean(arr_delay)))
    %>%
  mutate(n = n)
sf_25_means %>% fivenum_tbl(mean_arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  3.09  10.2 -20.1 -3.72   1.54     9  56.7
```

Vocabulary:

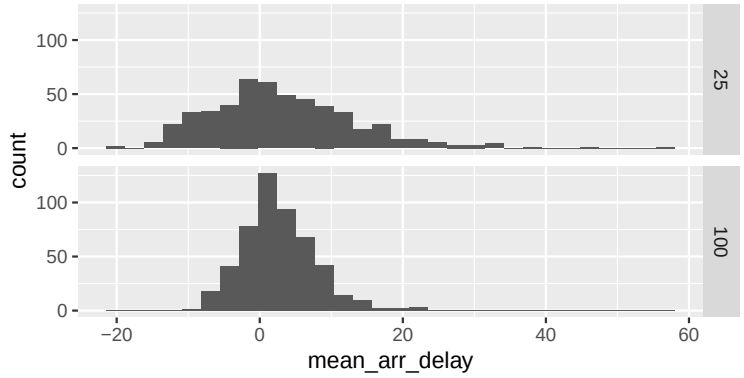
- the **sample size** n;
- the **sampling distribution**: the statistic over many trials;
- its **shape**: here somewhat right-skewed;
- the **standard error**: its standard deviation;
- the **95% confidence interval**: roughly mean ± 2 SE.

For $n = 25$ the means span about 75 minutes. With $n = 100$:

```
sf_100_means <- 1:num_trials %>%
  map_dfr(~ SF %>% slice_sample(n = 100) %>% summarize(mean_arr_delay = mean(arr_delay))
    ) %>%
  mutate(n = 100)
bind_rows(sf_25_means, sf_100_means) %>%
  group_by(n) %>%
  summarize(mean = mean(mean_arr_delay), se = sd(mean_arr_delay))
```

```
## # A tibble: 2 x 3
##       n  mean    se
##   <dbl> <dbl> <dbl>
## 1    25  3.09 10.2
## 2   100  2.58  4.99
```

```
bind_rows(sf_25_means, sf_100_means) %>%
  ggplot(aes(mean_arr_delay)) + geom_histogram(bins = 30) + facet_grid(n ~ .)
```



Quadrupling n halves the standard error and makes the distribution more symmetric.

The $1/\sqrt{n}$ law [Supplement]

For a sample mean of i.i.d. observations, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, so $\mathrm{SE} = \sigma/\sqrt{n}$: here $\sigma \approx 47.7$ gives 9.5 for $n = 25$ and 4.8 for $n = 100$, matching the simulation. (Without replacement multiply by $\sqrt{(N-n)/(N-1)} \approx 1$.) By the CLT the shape tends to normal, even for skewed delays.

4.4 The bootstrap

Motivation. With one sample, the bootstrap treats it as the population and resamples.

Resampling draws *with replacement* from the sample; duplication produces the variability. One sample of $n = 200$:

```
n <- 200
orig_sample <- SF %>% slice_sample(n = n, replace = FALSE)
sf_200_bs <- 1:num_trials %>%
  map_dfr(~ orig_sample %>% slice_sample(n = n, replace = TRUE) %>%
            summarize(mean_arr_delay = mean(arr_delay)))
sf_200_bs %>% fivenum_tbl(mean_arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  8.58  4.16 -1.94  5.47   8.42  11.3  23.7
```

```
sf_200_pop <- 1:num_trials %>%
  map_dfr(~ SF %>% slice_sample(n = n) %>% summarize(mean_arr_delay = mean(arr_delay)))
sf_200_pop %>% fivenum_tbl(mean_arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  2.69  3.32 -6.41 0.391   2.65  4.86  18.0
```

The **bootstrap distribution** (first table) is centred at the sample mean, not the population mean, but its sd approximates the true SE (second table); for moderate n it approximates standard errors and quantiles (Efron and Tibshirani, 1993). The 98th percentile and its bootstrap reliability:

```
orig_sample %>% summarize(q98 = quantile(arr_delay, p = 0.98))
```

```
## # A tibble: 1 x 1
##     q98
##   <dbl>
## 1  209.
```

```
sf_200_bs_q98 <- 1:num_trials %>%
  map_dfr(~ orig_sample %>% slice_sample(n = n, replace = TRUE) %>%
              summarize(q98 = quantile(arr_delay, p = 0.98)))
sf_200_bs_q98 %>% fivenum_tbl(q98)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  201.  58.2  74.1  152.    208  251.   360
```

The SE is large: extreme quantiles need far more data than means.

4.5 Outliers

Motivation. Extreme delays suggest which conditions make long delays likely.

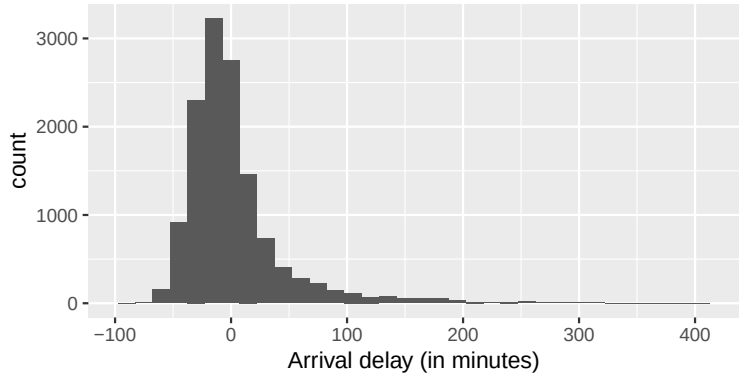
Delays of at least 420 minutes:

```
SF %>% filter(arr_delay >= 420) %>% select(month, day, dep_delay, arr_delay, carrier)
```

```
## # A tibble: 7 x 5
##   month   day dep_delay arr_delay carrier
##   <int> <int>     <dbl>     <dbl> <chr>
## 1    12     7       374       422 UA
## 2     7     6       589       561 DL
## 3     7     7       629       676 VX
## 4     7     7       653       632 VX
## 5     7    10       453       445 B6
## 6     7    10       432       433 VX
## # i 1 more row
```

Without them, most flights are on time or under an hour late.

```
SF %>% filter(arr_delay < 420) %>%
  ggplot(aes(arr_delay)) + geom_histogram(binwidth = 15) +
  labs(x = "Arrival delay (in minutes)")
```



Long delays (over an hour) by month and carrier:

```
SF %>% mutate(long_delay = arr_delay > 60) %>%
  count(month, long_delay) %>%
  pivot_wider(names_from = month, values_from = n)
```

```
## # A tibble: 2 x 13
##   long_delay   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <lgl>      <int> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
## 1 FALSE        856   741   812   993  1128   980   966  1159  1124  1177  1107  1093
## 2 TRUE          29    21    61   112    65   209   226    96    65    36    51    66
```

```
SF %>% mutate(long_delay = arr_delay > 60) %>%
  count(carrier, long_delay) %>%
  pivot_wider(names_from = carrier, values_from = n)
```

```
## # A tibble: 2 x 6
##   long_delay    AA    B6    DL    UA    VX
##   <lgl>      <int> <int> <int> <int> <int>
## 1 FALSE       1250   934  1757  6236  1959
## 2 TRUE         148    86    91   492   220
```

June and July are bad; Delta (DL) does well: arrive extra early in June–July, consider Delta.

4.6 Statistical models

Motivation. A model relates the response to several variables at once, separating signal from noise.

"Essentially, all models are wrong, but some are useful" (G. E. P. Box). We build models

1. to **understand** (which factors matter?), with simple interpretable models;
2. to **predict** unseen data, possibly with black boxes.

With response Y and predictors $X = (X_1, \ldots, X_p)$, we assume

$$Y = f(X) + \epsilon,$$

f unknown, $\mathbb{E}\epsilon = 0$. Estimate f for **prediction** $\hat{Y} = \hat{f}(X)$ or **inference** (which predictors matter, how, is linear enough).

Proposition 4.1 (reducible and irreducible error)

If $\hat{f}$ and X are fixed and $\mathbb{E}\epsilon = 0$, then

$$\mathbb{E}[(Y - \hat{Y})^2] = [f(X) - \hat{f}(X)]^2 + \mathrm{Var}(\epsilon).$$

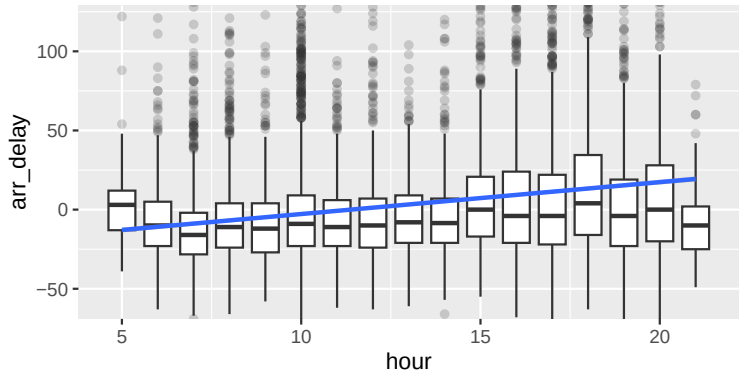
Proof [Supplement]. $Y - \hat{Y} = \{f(X) - \hat{f}(X)\} + \epsilon$, where the first term is a constant c. Hence $\mathbb{E}(Y - \hat{Y})^2 = c^2 + 2c\,\mathbb{E}\epsilon + \mathbb{E}\epsilon^2 = c^2 + \mathrm{Var}(\epsilon)$. ■

The first term (**reducible error**) shrinks with a better $\hat{f}$; the second (**irreducible error**) cannot.

Parametric models specify $f(X;\theta)$ with finitely many parameters, e.g. $Y \approx \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$; **nonparametric** models such as k-nearest neighbours assume no form for f. Flexibility costs interpretability.

Example. Do delays cascade through the day (SFO departures 5 am–9 pm)?

```
SF %>% ggplot(aes(hour, arr_delay)) +
  geom_boxplot(aes(group = hour), outlier.alpha = 0.2) +
  geom_smooth(method = "lm", formula = y ~ x) +
  coord_cartesian(ylim = c(-60, 120))
```




```
mod1 <- lm(arr_delay ~ hour, data = SF)
broom::tidy(mod1)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)   -22.9     1.23       -18.6 2.88e- 76
## 2 hour            2.01    0.0915      22.0 1.78e-105
```

Mean delay rises about 2 minutes per hour; the tiny p-value rules out no relationship.

4.7 Selected exercises

Motivation. Exact counterparts of the lecture's simulations.

Exercise 4.1 (what a resample contains)

A bootstrap resample draws n times with replacement from $x_1, \dots, x_n$.
(a) Show that a fixed observation x_i is absent from the resample with probability $(1-1/n)^n$, and that this tends to $e^{-1} \approx 0.368$.
(b) Show that the expected number of distinct original observations in a resample is $n\{1-(1-1/n)^n\} \approx 0.632\,n$.

Solution [Supplement].

(a) Each of n independent draws misses i with probability $1-1/n$, so $P(x_i \text{ absent}) = (1-1/n)^n$. With $t = 1/n \to 0$, $n\log(1-1/n) = \log(1-t)/t \to -1$ (derivative of $\log(1-t)$ at 0), so the limit is e^{-1} (0.367 for $n = 200$).

(b) With $I_i = 1\{x_i \text{ appears}\}$, by linearity and (a), $\mathbb{E}\sum_i I_i = \sum_i P(I_i = 1) = n\{1-(1-1/n)^n\} \approx (1-e^{-1})n = 0.632\,n$. ■

Exercise 4.2 (bootstrap standard error of a mean)

Condition on the data and let $\bar{x}^*$ be the mean of a bootstrap resample. Show that $\mathbb{E}^*\bar{x}^* = \bar{x}$ and $\mathrm{Var}^*\,\bar{x}^* = \hat{\sigma}^2/n$ with $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$, where $\mathbb{E}^*$, Var^* refer to resampling only. Compare with the usual estimate $s/\sqrt{n}$ of the standard error.

Solution [Supplement].

Step 1. The resampled values $x_1^*, \dots, x_n^*$ are i.i.d., each uniform on the multiset $\{x_1, \dots, x_n\}$. Hence $\mathbb{E}^* x_j^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\mathrm{Var}^*\, x_j^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$.

Step 2. By linearity, $\mathbb{E}^*\bar{x}^* = \bar{x}$ (centred at the sample mean).

Step 3. By independence, $\operatorname{Var}^* \bar{x}^* = \frac{1}{n^2}\sum_j \operatorname{Var}^* x_j^* = \hat{\sigma}^2/n$.

Step 4. Since $\hat{\sigma}^2 = \frac{n-1}{n}s^2$, the bootstrap SE is $\sqrt{(n-1)/n}\; s/\sqrt{n}$, i.e. $s/\sqrt{n}$ times 0.9975 for $n = 200$; simulation adds Monte Carlo error. ■

Exercise 4.3 (the finite-population correction)

A population has values $v_1, \dots, v_N$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum_j (v_j - \mu)^2$. A simple random sample $X_1, \dots, X_n$ is drawn without replacement ($n \le N$). Prove that

$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Solution [Supplement].

Step 1. By symmetry X_i is uniform on the population ($\mathbb{E}X_i = \mu$, $\operatorname{Var} X_i = \sigma^2$), and for $i \neq j$, (X_i, X_j) is uniform on ordered pairs of distinct units, so $c = \operatorname{Cov}(X_i, X_j)$ does not depend on i, j, n.

Step 2. Find c from the full census $n = N$: then $\sum_{i=1}^N X_i = \sum_j v_j$ is constant, so

$$0 = \operatorname{Var}\Big(\sum_{i=1}^N X_i\Big) = N\sigma^2 + N(N-1)c, \qquad c = -\frac{\sigma^2}{N-1}.$$

Step 3. For general n,

$$\operatorname{Var}(\bar{X}) = \frac{1}{n^2}\{n\sigma^2 + n(n-1)c\} = \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

The factor is ≈ 1 for $n \ll N$ and 0 for a census. ■

Exercise 4.4 (how large a sample)

For the SFO arrival delays, $\sigma \approx 47.7$ minutes. How many flights are needed for the 95% confidence interval $\bar{X} \pm 1.96\,\sigma/\sqrt{n}$ for the mean delay to have half-width at most 5 minutes? By what factor must n grow to halve the half-width again?

Solution [Supplement]. $1.96\,\sigma/\sqrt{n} \le 5 \iff n \ge (1.96 \cdot 47.7/5)^2 = 18.698^2 = 349.6$, so $n = 350$. Half-width $\propto 1/\sqrt{n}$, so halving it needs $4n = 1400$. ■

Lecture 5. Linear Regression

5.1 Simple linear regression

Motivation. Linear regression is the simplest complete answer to how Y depends on predictors; its ideas carry over to later models.

Estimation fits parameters; **prediction** applies the fit to unseen data. y is the *outcome*, x the *predictor*. The model:

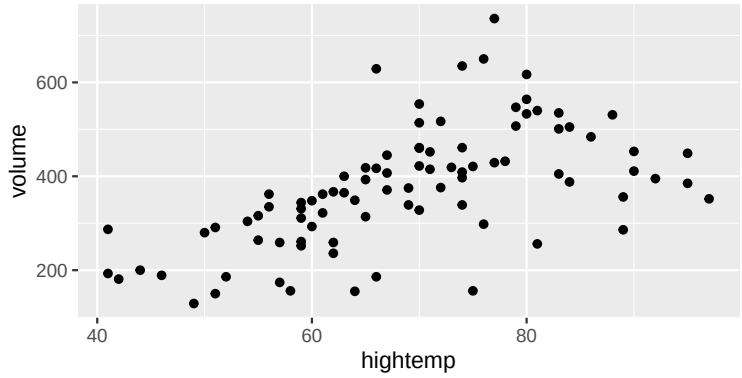
$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad i = 1, \dots, n,$$

with $\mathbb{E}\epsilon_i = 0$. `lm()` gives $\hat\beta_0$, $\hat\beta_1$, fitted values $\hat y_i = \hat\beta_0 + \hat\beta_1 x_i$ and residuals $\hat\epsilon_i = y_i - \hat y_i$.

5.1.1 Example: the rail trail

`mosaicData::RailTrail`: daily `volume` of trail users in Florence, Massachusetts, over 90 days, with temperature, rain, cloud cover and weekday.

```
ggplot(RailTrail, aes(x = hightemp, y = volume)) + geom_point()
```




```
mod <- lm(volume ~ hightemp, data = RailTrail)
tidy(mod)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic       p.value
##   <chr>          <dbl>     <dbl>     <dbl>         <dbl>
## 1 (Intercept)   -17.1     59.4      -0.288 0.774
## 2 hightemp        5.70     0.848     6.72  0.00000000171
```

$\hat\beta_1 = 5.70$ users per °F; the intercept (0°F) is outside the data. `summary(mod)` adds residual SE, R^2, F-test:

```
summary(mod)
```

```
##
## Call:
## lm(formula = volume ~ hightemp, data = RailTrail)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -254.562  -57.800    8.737   57.352  314.035
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -17.079     59.395  -0.288    0.774
## hightemp       5.702      0.848   6.724 1.71e-09 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 104.2 on 88 degrees of freedom
## Multiple R-squared:  0.3394, Adjusted R-squared:  0.3319
## F-statistic: 45.21 on 1 and 88 DF,  p-value: 1.705e-09
```

5.1.2 Prediction and intervals

```
predict(mod, newdata = data.frame(hightemp = c(30, 40)))
```

```
##        1        2
## 153.9771 210.9958
```

```
predict(mod, interval = "confidence")[1:3, ]
```

```
##        fit      lwr      upr
## 1 456.1766 423.8301 488.5231
## 2 399.1578 376.2313 422.0843
## 3 404.8597 381.3622 428.3572
```

```
predict(mod, interval = "prediction")[1:3, ]
```

```
## Warning in predict.lm(mod, interval = "prediction"): predictions on current data refer to
    _future_ responses
```

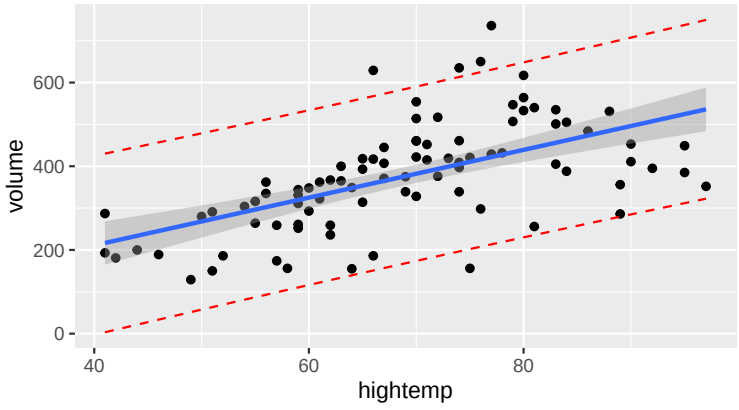
```
##        fit      lwr      upr
## 1 456.1766 246.6175 665.7357
## 2 399.1578 190.8447 607.4709
## 3 404.8597 196.4830 613.2364
```

Confidence interval: mean volume at x; **prediction interval**: one new day, wider by the noise ϵ.

```
pi <- predict(mod, interval = "prediction")
```

```
## Warning in predict.lm(mod, interval = "prediction"): predictions on current data refer to
    _future_ responses
```

```
RailTrail %>%
  mutate(lower.pi = pi[, "lwr"], upper.pi = pi[, "upr"]) %>%
  ggplot(aes(x = hightemp, y = volume)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE) +
  geom_line(aes(y = lower.pi), colour = "red", linetype = "dashed") +
  geom_line(aes(y = upper.pi), colour = "red", linetype = "dashed")
```



The two interval formulas [Supplement]

With $S_{xx} = \sum(x_i - \bar{x})^2$ and $s^2 = \text{SSE}/(n-2)$, the intervals at x_0 are

$$\hat{y}_0 \pm t_{n-2,0.975}\, s\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}} \quad \text{(mean)}, \qquad \hat{y}_0 \pm t_{n-2,0.975}\, s\sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}} \quad \text{(new obs.)}.$$

The extra 1 is the new error's variance; both widen away from $\bar{x}$.

5.1.3 Strength of fit and least squares

Beyond the correlation r:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \in [0,1],$$

the fraction of variation explained relative to the null model $y_i = \beta_0 + \epsilon_i$. Least squares minimises SSE:

$$\sum_i \{y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)\}^2 \le \sum_i \{y_i - (\beta_0 + \beta_1 x_i)\}^2 \quad \text{for all } \beta_0, \beta_1.$$

Least squares is computationally convenient but, like the mean, outlier-sensitive. Generally, *a statistical method optimises a goodness-of-fit measure over a family of models*, by formula or by algorithm.

5.1.4 A categorical predictor

With the indicator `weekday`, $\text{volume}_i = \beta_0 + \beta_1 \text{weekday}_i + \epsilon_i$ fits the two group means:

```
RailTrail %>% group_by(weekday) %>% summarise(mean_volume = mean(volume))
```

```
## # A tibble: 2 x 2
##   weekday mean_volume
##   <lgl>         <dbl>
## 1 FALSE          431.
## 2 TRUE           350.
```

```
coef(lm(volume ~ weekday, data = RailTrail))
```

```
## (Intercept) weekdayTRUE
##   430.71429   -80.29493
```

About 80 fewer riders on weekdays. The alphabetically first level is the **reference group**; recoding flips the sign:

```
RailTrail <- RailTrail %>%
  mutate(day = ifelse(weekday == 1, "weekday", "weekend/holiday"))
coef(lm(volume ~ day, data = RailTrail))
```

```
##        (Intercept) dayweekend/holiday
##          350.41935           80.29493
```

5.2 Loss functions

Motivation. The loss function defines "best fit"; different losses differ most with outliers.

A **loss function** $L(y, \hat{y})$: squared (L_2) $(y-\hat{y})^2$ or absolute (L_1) $|y-\hat{y}|$; L_2 penalises large residuals more. Fitting minimises the **empirical risk**

$$\hat{R}(\beta_0, \beta_1) = \frac{1}{n}\sum_i L(y_i, \hat{y}_i),$$

(MSE for L_2, MAE for L_1). Least squares minimises the L_2 risk; zero derivatives give the estimating equations $\frac{1}{n}\sum_i (y_i - \hat{y}_i) = 0$ and $\frac{1}{n}\sum_i (y_i - \hat{y}_i)x_i = 0$, whose solution is

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

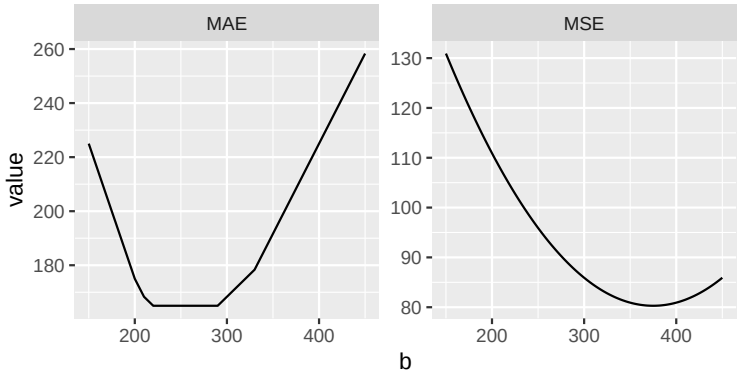
Proposition 5.1 (constant model)

For the constant model $y = \beta_0 + \epsilon$, the minimiser of the L_2 risk is $\bar{y}$ and a minimiser of the L_1 risk is any median of $y_1, \ldots, y_n$.

Proof. L_2: $\frac{d}{d\beta_0}\frac{1}{n}\sum(y_i - \beta_0)^2 = -\frac{2}{n}\sum(y_i - \beta_0) = 0$ gives $\beta_0 = \bar{y}$, and the risk is a convex parabola. L_1: away from the data points, $\frac{d}{d\beta_0}\frac{1}{n}\sum|y_i - \beta_0| = \frac{1}{n}(\#\{y_i < \beta_0\} - \#\{y_i > \beta_0\})$. The risk is convex and piecewise linear, decreasing while fewer than half the points lie below β_0 and increasing once more than half do, so it is minimised at a median. ■

Example (bubble-tea sales). Sales 200, 210, 220, 290, 330: mean 250 (L_2), median 220 (L_1). Adding 1000 moves the mean to 375, the median to 255; with six points any value in $[220, 290]$ minimises the MAE.

```
y <- c(200, 210, 220, 290, 330, 1000)
tibble(b = seq(150, 450, by = 1)) %>%
  mutate(MSE = map_dbl(b, ~ mean((y - .x)^2)) / 1000,
         MAE = map_dbl(b, ~ mean(abs(y - .x)))) %>%
  pivot_longer(-b) %>%
  ggplot(aes(b, value)) + geom_line() + facet_wrap(~ name, scales = "free_y")
```



MSE (in thousands): smooth, unique minimum, outlier-sensitive. MAE: piecewise linear, flat on $[220, 290]$, robust.

5.3 Multiple regression

Motivation. Multiple regression estimates each effect holding the others fixed.

$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$; β_j is the change in y per unit of x_j *holding the others fixed.*

```
mod_plane <- lm(volume ~ hightemp + precip, data = RailTrail)
tidy(mod_plane)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   -31.5     55.2      -0.571 5.70e- 1
## 2 hightemp        6.12     0.794     7.70  1.97e-11
## 3 precip       -153.      39.3      -3.90  1.90e- 4
```

An inch of rain lowers ridership by about 153 at fixed temperature.

5.3.1 Inference for coefficients

Normal-approximation 95% interval: $\hat{\beta}_j \pm 1.96\,\mathrm{SE}(\hat{\beta}_j)$; `confint()` uses t quantiles:

```
mod_p_planes <- lm(volume ~ hightemp + precip + weekday, data = RailTrail)
tidy(mod_p_planes)
```

```
## # A tibble: 4 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    19.3     60.3       0.320 7.50e- 1
## 2 hightemp        5.80     0.799     7.26  1.59e-10
## 3 precip       -146.      38.9      -3.74  3.27e- 4
## 4 weekdayTRUE   -43.1     22.2      -1.94  5.52e- 2
```

```
confint(mod_p_planes)
```

```
##                   2.5 %      97.5 %
## (Intercept) -100.630740 139.2683959
## hightemp       4.213254   7.3880945
## precip      -222.927105 -68.2908651
## weekdayTRUE  -87.265016   0.9763902
```

Bootstrap (Lecture 4): resample days, refit, take the sd of the coefficients.

```
set.seed(1)
n <- nrow(RailTrail)
coef_fun <- function(data) lm(volume ~ hightemp + precip + weekday, data = data)$coef
boot <- 1:500 %>%
  map_dfr(~ RailTrail %>% slice_sample(n = n, replace = TRUE) %>% coef_fun())
round(coef(mod_p_planes)["hightemp"] + c(-1.96, 1.96) * sd(boot$hightemp), 2)
```

```
## [1] 4.2 7.4
```

Close to the model-based interval.

5.3.2 Choosing a model

R^2 always grows with more variables; **Occam's razor** prefers the simpler model. Candidates (`volume ~ .` uses all):

```
mod1 <- lm(volume ~ hightemp + weekday, data = RailTrail)
mod2 <- lm(volume ~ hightemp + cloudcover + weekday, data = RailTrail)
mod3 <- lm(volume ~ . - day, data = RailTrail)
map_dfr(list(mod1 = mod1, mod2 = mod2, mod3 = mod3), glance, .id = "model") %>%
  select(model, r.squared, adj.r.squared, AIC, BIC)
```

```
## # A tibble: 3 x 5
##   model r.squared adj.r.squared   AIC   BIC
##   <chr>     <dbl>         <dbl> <dbl> <dbl>
## 1 mod1      0.374         0.359 1093. 1103.
## 2 mod2      0.459         0.440 1082. 1094.
```

```
## 3 mod3      0.511         0.469 1081. 1103.
```

BIC prefers `mod2`; adjusted R^2 and AIC slightly favour the full model. Criteria penalising size:

- adjusted $R^2_{\text{adj}} = 1 - (1 - R^2)\dfrac{n-1}{n-p-1}$, with p the number of predictors;
- AIC (Akaike's information criterion) and BIC (Bayesian, with a stronger penalty on p), smaller is better;
- Mallows' C_p.

All-subset regression searches 2^p models; **stepwise regression** goes backward, forward or both (`MASS::stepAIC()`); **penalized regression** (Lecture 7) shrinks coefficients. In-sample selection can **overfit**; cross-validation (Lecture 6) guards against it, essential for flexible methods.

5.4 Other considerations

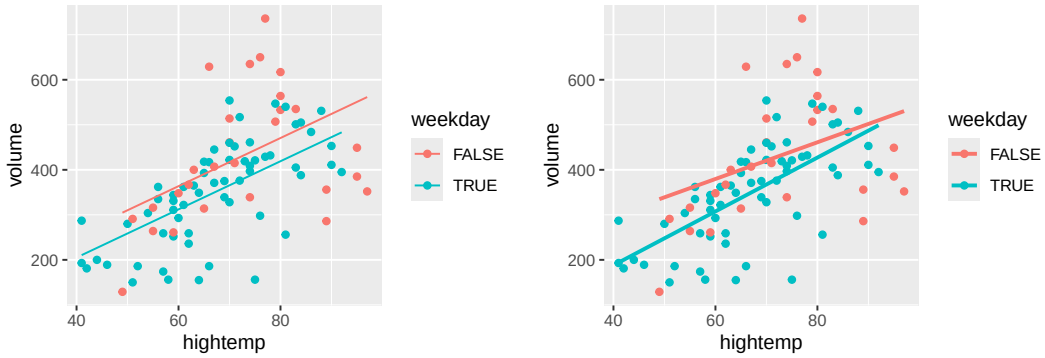
Factor variables are coded by dummy (indicator) variables: *reference coding* drops one level, *one-hot encoding* keeps all (and then drops the intercept). Ordered categories may be treated as numeric scores.

Interactions. The linear model is *additive* and *linear*. With 0/1 x_2 (weekday), $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ gives parallel lines (intercepts $\hat{\beta}_0$, $\hat{\beta}_0 + \hat{\beta}_2$). Adding $x_1 x_2$,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2,$$

gives slope $\hat{\beta}_1$ on weekends and $\hat{\beta}_1 + \hat{\beta}_3$ on weekdays: an **interaction model**.

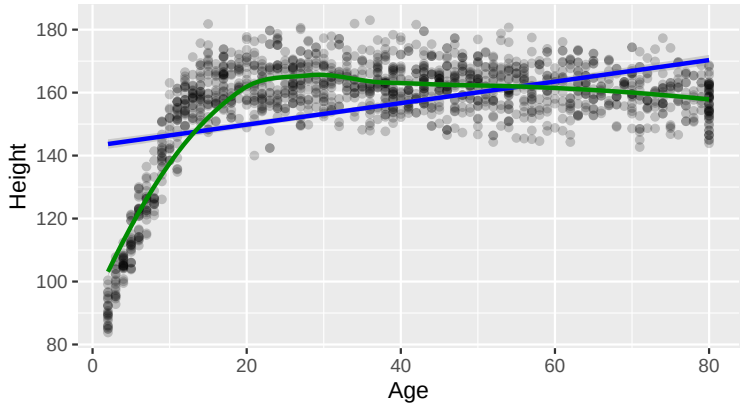
```
ggplot(RailTrail, aes(hightemp, volume, colour = weekday)) +
  geom_point() +
  geom_line(aes(y = fitted(lm(volume ~ hightemp + weekday, data = RailTrail))))
ggplot(RailTrail, aes(hightemp, volume, colour = weekday)) +
  geom_point() + geom_smooth(method = "lm", formula = y ~ x, se = FALSE)
```



Left: parallel slopes; right: separate slopes (interaction).

Nonlinear relationships. Female height against age (`NHANES`): a line misfits, LOESS follows the growth curve with a wider band (Lecture 10).

```
NHANES %>% filter(Gender == "female", !is.na(Height)) %>%
  slice_sample(n = 1500) %>%
  ggplot(aes(Age, Height)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", formula = y ~ x, colour = "blue") +
  geom_smooth(method = "loess", formula = y ~ x, colour = "green4")
```



5.5 Selected exercises

Motivation. Least-squares facts used without proof, each from the estimating equations.

Exercise 5.1 (R^2 in simple regression)

For the least-squares line with residuals $e_i = y_i - \hat{y}_i$, and assuming SST > 0:
(a) show that $\sum_i e_i = 0$, $\sum_i x_i e_i = 0$ and SST $=$ SSR $+$ SSE with SSR $= \sum_i (\hat{y}_i - \bar{y})^2$; conclude $0 \le R^2 \le 1$;
(b) show that $R^2 = r^2$, the squared sample correlation of x and y.

Solution [Supplement].

(a) The first two identities are the estimating equations of Section 5.2, multiplied by n. Since $\hat{y}_i - \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i - \bar{y}$ is a linear combination of 1 and x_i, they give $\sum_i (\hat{y}_i - \bar{y}) e_i = 0$. Expanding $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + e_i$,

$$\text{SST} = \text{SSR} + \text{SSE} + 2\sum_i (\hat{y}_i - \bar{y}) e_i = \text{SSR} + \text{SSE}.$$

Both terms are ≥ 0, so $R^2 = 1 - \text{SSE/SST} = \text{SSR/SST} \in [0, 1]$.

(b) Write $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$ and $S_{yy} = \text{SST}$. From $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ we get $\hat{y}_i - \bar{y} = \hat{\beta}_1 (x_i - \bar{x})$, so

$$\text{SSR} = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}, \qquad R^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r^2.$$

■

Exercise 5.2 (indicators reproduce group means)

(a) For $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $x_i \in \{0, 1\}$ and both groups non-empty, show that $\hat{\beta}_0 = \bar{y}_0$ and $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$, the group means and their difference.
(b) For the interaction model $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon$ with $x_2 \in \{0, 1\}$, show that the least-squares fit equals the two simple regressions of y on x_1 fitted separately in the groups $x_2 = 0$ and $x_2 = 1$ (assume x_1 is not constant in either group).

Solution [Supplement].

(a) Put $a = \beta_0$ and $b = \beta_0 + \beta_1$; the map $(\beta_0, \beta_1) \mapsto (a, b)$ is a bijection. The residual sum of squares splits as

$$\sum_{i: x_i = 0} (y_i - a)^2 + \sum_{i: x_i = 1} (y_i - b)^2,$$

each minimised separately; by Proposition 5.1 the minimisers are $\hat{a} = \bar{y}_0$ and $\hat{b} = \bar{y}_1$, hence $\hat{\beta}_0 = \bar{y}_0$ and $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$.

(b) Put $(a_0, a_1) = (\beta_0, \beta_1)$ and $(b_0, b_1) = (\beta_0 + \beta_2, \beta_1 + \beta_3)$, again a bijection. The fitted value is $a_0 + a_1 x_1$ when $x_2 = 0$ and $b_0 + b_1 x_1$ when $x_2 = 1$, so

$$\mathrm{RSS} = \sum_{i : x_{i2} = 0} (y_i - a_0 - a_1 x_{i1})^2 + \sum_{i : x_{i2} = 1} (y_i - b_0 - b_1 x_{i1})^2.$$

Each sum is a within-group simple-regression RSS in its own parameters, minimised by the separate least-squares lines (unique since x_1 varies). The interaction model adds a single model in which β_3, the slope difference, can be tested. ■

Exercise 5.3 (precision of the slope)

Treat $x_1, \dots, x_n$ as fixed and let the errors be uncorrelated with mean 0 and variance σ^2.
(a) Show that $\hat{\beta}_1$ is unbiased with $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$.
(b) If the x_i must lie in $[L, U]$, show that $S_{xx} \le n(U - L)^2/4$, with equality when half of the points (for even n) are at each end. What does this say about designing a study?

Solution [Supplement].

(a) Because $\sum_i (x_i - \bar{x}) = 0$, $\hat{\beta}_1 = \sum_i (x_i - \bar{x}) y_i / S_{xx}$, a linear combination of the y_i with weights $w_i = (x_i - \bar{x})/S_{xx}$. Then

$$\mathbb{E}\hat{\beta}_1 = \sum_i w_i (\beta_0 + \beta_1 x_i) = \beta_1 \frac{\sum_i (x_i - \bar{x}) x_i}{S_{xx}} = \beta_1,$$

since $\sum_i w_i = 0$ and $\sum_i (x_i - \bar{x}) x_i = S_{xx}$. With uncorrelated errors, $\mathrm{Var}\,\hat{\beta}_1 = \sigma^2 \sum_i w_i^2 = \sigma^2 / S_{xx}$.

(b) Let $m = (L + U)/2$. The mean minimises the sum of squared deviations (Proposition 5.1), so $S_{xx} \le \sum_i (x_i - m)^2 \le n\{(U - L)/2\}^2$, because $|x_i - m| \le (U - L)/2$. With half the points at each end, $\bar{x} = m$ and $|x_i - m| = (U - L)/2$, so equality holds. Spreading x maximises precision, but some interior points are needed to check linearity. ■

Lecture 6. Generalized Linear Models and Model Evaluation

6.1 Generalized linear models

Motivation. GLMs link a linear predictor to the mean of a binary, count or other restricted outcome.

A monotone **link function** f:

$$f(\theta_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}, \qquad y_i \sim D(\theta_i, \psi),$$

where θ_i is the centre of the distribution D and ψ controls its scale or shape.

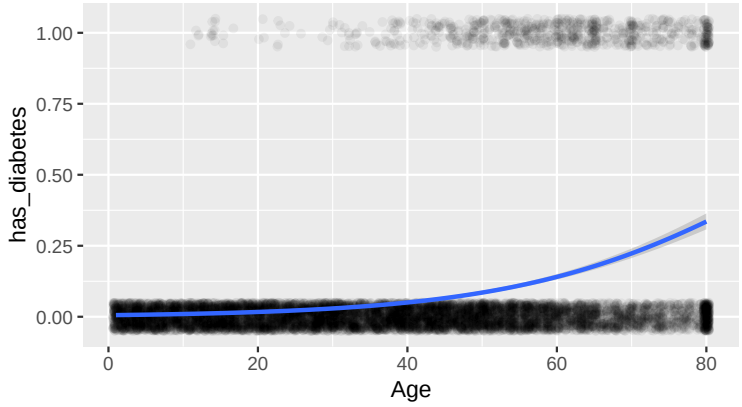
6.2 Logistic regression

For binary $y \sim \text{Bernoulli}(\theta)$ the link is the **logit**:

$$\text{logit}(\theta) = \log \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \theta = \frac{1}{1 + \exp\{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)\}} \in (0,1).$$

Diabetes against age and BMI (`NHANES`):

```
NHANES <- NHANES %>% mutate(has_diabetes = as.numeric(Diabetes == "Yes"))
NHANES %>% filter(!is.na(has_diabetes)) %>%
  ggplot(aes(Age, has_diabetes)) +
  geom_jitter(height = 0.05, alpha = 0.05) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"), formula = y ~ x)
```



The probability rises with age (and BMI). Both together:

```
logreg <- glm(has_diabetes ~ BMI + Age, family = "binomial", data = NHANES)
tidy(logreg)
```

```
## # A tibble: 3 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)  -8.08     0.244       -33.1 1.30e-239
## 2 BMI           0.0943   0.00552      17.1 1.74e- 65
## 3 Age           0.0573   0.00249      23.0 2.28e-117
```

Predicted probability at age 60, BMI 25:

```r
linear_component <- c(1, 25, 60) %*% coef(logreg)
exp(linear_component) / (1 + exp(linear_component))
```

```
##             [,1]
## [1,] 0.09233095
```

```r
predict(logreg, newdata = data.frame(BMI = 25, Age = 60), type = "response")
```

```
##          1
## 0.09233095
```

6.2.1 Coefficients as odds ratios

A unit change in x does not change the probability by a constant amount, so coefficients are read through **odds** $p/(1-p)$ and **odds ratios**. With a single 0/1 predictor, logit $P(y_i = 1) = \beta_0 + \beta_1 x_i$, the log odds are $\beta_0 + \beta_1$ when $x_i = 1$ and β_0 when $x_i = 0$, so with $p = P(y = 1 \mid x = 1)$ and $q = P(y = 1 \mid x = 0)$,

$$e^{\beta_1} = \frac{p/(1-p)}{q/(1-q)}.$$

In general e^{β_j} multiplies the odds per unit of x_j, others fixed:

```r
exp(coef(logreg))
```

```
##  (Intercept)          BMI          Age
## 0.0003095805 1.0989173575 1.0589497173
```

Odds $\times 1.06$ per year of age, $\times 1.10$ per BMI unit.

6.3 GLMs for counts

Motivation. Count variance grows with the mean: Poisson, then negative binomial and zero-inflated models.

6.3.1 Poisson regression

A Poisson variable with rate λ has $P(x = k \mid \lambda) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$, mean = variance $= \lambda$. Poisson regression: $y_i \sim \text{Poisson}(\lambda_i)$ with the log link,

$$\log \lambda_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip},$$

no ψ; e^{β_j} multiplies the rate: $\lambda(x_j + 1) = \lambda(x_j)\, e^{\beta_j}$.

Example. `pscl::bioChemists`: articles (`art`) of 915 biochemistry PhD students in the last three PhD years, with gender, marriage, kids under 5, prestige, mentor's articles.

```r
bioChemists <- bioChemists %>% mutate(gender = ifelse(fem == "Men", 1, 0))
pois_mod <- glm(art ~ gender, family = "poisson", data = bioChemists)
round(coef(pois_mod), 3)
```

```
## (Intercept)      gender
##       0.385       0.247
```

```r
exp(coef(pois_mod))
```

```
## (Intercept)      gender
##    1.470309    1.280405
```

Women: $\hat{\lambda} = e^{0.385} = 1.47$; men: $e^{0.385+0.247} = 1.88$, ratio $e^{0.247} = 1.28$. (The slides write $\lambda_{\text{women}} = e^{0.247}$; the women's rate is $e^{\hat{\beta}_0} = e^{0.385}$, which is the value 1.470 given there [Supplement].)

6.3.2 Overdispersion and the negative binomial

Overdispersion: variance above the model's (for counts, above the mean); a Poisson fit then *underestimates* standard errors.

```
var(bioChemists$art) / mean(bioChemists$art)
```

```
## [1] 2.191358
```

Variance over twice the mean. The **negative binomial** counts failures before the r-th success:

$$P(x = k \mid r, \theta) = \binom{r+k-1}{k} \theta^r (1-\theta)^k = \frac{\Gamma(r+k)}{\Gamma(r)\, k!} \theta^r (1-\theta)^k,$$

(Gamma form: non-integer r.) `dnbinom(2, 3, 0.75)` $= P(x = 2 \mid r = 3, \theta = 0.75)$. Mean $\mu = r(1-\theta)/\theta$, so $\theta = r/(r+\mu)$; variance $\mu + \mu^2/r$.

```
dnbinom(2, 3, 0.75)
```

```
## [1] 0.1582031
```

It is a Poisson mixture with gamma-distributed means.

Poisson–gamma mixture [Supplement]

If $\lambda \sim$ Gamma(shape r, rate r/μ) and $x \mid \lambda \sim$ Poisson(λ), then

$$P(x = k) = \int_0^\infty \frac{e^{-\lambda}\lambda^k}{k!} \frac{(r/\mu)^r \lambda^{r-1} e^{-r\lambda/\mu}}{\Gamma(r)}\, d\lambda = \frac{\Gamma(r+k)}{\Gamma(r)\, k!} \Big(\frac{r}{r+\mu}\Big)^r \Big(\frac{\mu}{r+\mu}\Big)^k,$$

the negative binomial with $\theta = r/(r+\mu)$; μ^2/r is the variance of the rates.

Negative binomial regression assumes $y_i \sim \text{NegBin}(\mu_i, r)$ with $\log \mu_i = \beta_0 + \beta x_i$; `MASS::glm.nb()` fits it:

```
nb_mod <- glm.nb(art ~ gender, data = bioChemists)
tidy(nb_mod)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    0.385    0.0544      7.08 1.44e-12
## 2 gender         0.247    0.0720      3.43 6.01e- 4
```

```
tidy(pois_mod)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)    0.385    0.0402      9.59 8.77e-22
## 2 gender         0.247    0.0519      4.77 1.89e- 6
```

Same coefficients, larger SEs: Poisson was overconfident.

6.3.3 Zero-inflated Poisson

Excess zeros (cigarettes smoked: non-smokers report 0). **Zero-inflated Poisson** (ZIP): with probability θ_i a "structural zero" ($z_i = 1$), otherwise Poisson,

$$y_i \sim \theta_i \mathbb{1}(y_i = 0) + (1-\theta_i)\,\text{Poisson}(\lambda_i), \qquad \log \lambda_i = \beta_0 + \beta_1 x_i, \quad \text{logit}\, \theta_i = \gamma_0 + \gamma_1 x_i,$$

Slides (`zeroinfl(cigs ~ educ)`): more education, more non-smokers ($\hat\theta$ 0.49 at 6 years to 0.72 at 18) but heavier smokers ($\hat\lambda$ 18.3 to 27.7). With `bioChemists`:

```
zip_mod <- zeroinfl(art ~ gender | gender, data = bioChemists)
zip_mod
```

```
##
## Call:
## zeroinfl(formula = art ~ gender | gender, data = bioChemists)
##
## Count model coefficients (poisson with log link):
## (Intercept)       gender
##      0.6151       0.2393
##
## Zero-inflation model coefficients (binomial with logit link):
## (Intercept)       gender
##    -1.35416     -0.03902
```

Count part: men publish more when they publish; zero part: gender effect on being a "non-publisher".

6.4 Evaluating models

Motivation. Training fit is optimistic; honest evaluation needs unseen data.

We seek g with $y = g(x) + \epsilon$ (linear: $g(x) = \beta_0 + \cdots + \beta_p x_p$; GLM: $g(x) = f^{-1}(\beta_0 + \cdots + \beta_p x_p)$).

A coin example. Estimating a fair coin's p: a uniform random guess has zero bias, large variance; always 0.7 has zero variance, large bias.

Proposition 6.1 (bias–variance decomposition)

For a new point (x, y) with $y = g(x) + \epsilon$, $\mathbb{E}\epsilon = 0$, and ϵ independent of the fitted model $\hat{y}(\cdot)$,

$$\mathbb{E}\big[(y - \hat{y}(x))^2\big] = \mathbb{E}[\epsilon^2] + \big(g(x) - \mathbb{E}[\hat{y}(x)]\big)^2 + \mathbb{E}\big[(\mathbb{E}[\hat{y}(x)] - \hat{y}(x))^2\big].$$

Proof [Supplement]. $y - \hat{y} = \epsilon + \{g(x) - \mathbb{E}\hat{y}\} + \{\mathbb{E}\hat{y} - \hat{y}\}$; square and take expectations. The middle term is constant; $\mathbb{E}\epsilon = 0$ and independence kill the cross terms with ϵ; $\mathbb{E}\{\mathbb{E}\hat{y} - \hat{y}\} = 0$ kills the last one. ■

Model risk = observation variance + (model bias)^2 + model variance.

- *Observation variance* σ^2: irreducible noise.
- *Model bias*: too limited a family (quadratic for cubic data), **underfitting**; reduced by complexity.
- *Model variance*: fit changes a lot across samples, **overfitting**; reduced by simplicity.

With complexity, bias falls and variance rises: risk is U-shaped.

6.4.1 Training error and the holdout method

Training error favours complexity. **Holdout**: set data aside. `RailTrail`: hold out every fifth day (18 of 90), fit degrees 1–5 on 72:

```
index <- (1:18) * 5
test <- RailTrail[index, ]
train <- RailTrail[-index, ]
mods <- map(1:5, ~ glm(volume ~ poly(hightemp, .x), data = train))
tibble(degree = 1:5,
```

```
        train_mse = map_dbl(mods, ~ mean((train$volume - predict(.x))^2)),
        valid_mse = map_dbl(mods, ~ mean((test$volume - predict(.x, newdata = test))^2)),
        AIC = map_dbl(mods, AIC))
```

```
## # A tibble: 5 x 4
##   degree train_mse valid_mse   AIC
##    <int>     <dbl>     <dbl> <dbl>
## 1      1    11429.     7835.  883.
## 2      2    10277.     6531.  877.
## 3      3     9318.     4975.  872.
## 4      4     9210.     5128.  874.
## 5      5     9203.     5023.  875.
```

Training error keeps falling; validation error and AIC both pick degree 3. The held-out part is a *validation* set; splits 80/20, 90/10, 75/25 are common.

6.4.2 Cross-validation

A single split wastes data. **k-fold cross-validation**: random k folds, each validating a fit on the other $k-1$, giving $\mathrm{MSE}_1, \ldots, \mathrm{MSE}_k$, and

$$\mathrm{CV}_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i.$$

Leave-one-out CV ($k = n$): no randomness, less bias than holdout, n fits. `boot::cv.glm()`:

```
set.seed(1)
cv_error <- map_dbl(1:5, function(i) {
  fit <- glm(volume ~ poly(hightemp, i), data = RailTrail)
  boot::cv.glm(RailTrail, fit, K = 5)$delta[1]
})
cv_error
```

```
## [1] 11386.289  9845.353  8797.032  9319.328  9336.853
```

LOOCV without refitting [Supplement]

For least squares, leaving out observation i changes its residual to $e_i/(1-h_{ii})$, with h_{ii} the leverage, so $\mathrm{CV}_{(n)} = \frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ needs a single fit.

```
fit <- lm(volume ~ poly(hightemp, 3), data = RailTrail)
mean((residuals(fit) / (1 - hatvalues(fit)))^2)
```

```
## [1] 8992.758
```

```
boot::cv.glm(RailTrail, glm(volume ~ poly(hightemp, 3), data = RailTrail))$delta[1]
```

```
## [1] 8992.758
```

6.4.3 Why a separate test set

The best of many validation losses is optimistic (a tournament winner was also lucky). Report a **test set** never used for fitting or selection: train to fit, validate (or CV) to select, test once.

6.5 Selected exercises

Motivation. Three results behind this lecture's numbers.

Exercise 6.1 (one binary predictor)

Let $x_i \in \{0,1\}$, with n_g observations and mean response $\bar{y}_g$ in group $g = 0, 1$.
(a) In the logistic regression logit $P(y_i = 1) = \beta_0 + \beta_1 x_i$, suppose $0 < \bar{y}_0, \bar{y}_1 < 1$. Show that the maximum likelihood fit gives $\hat{P}(y = 1 \mid x = g) = \bar{y}_g$, so $e^{\hat{\beta}_1}$ is the sample odds ratio.
(b) In the Poisson regression $\log \lambda_i = \beta_0 + \beta_1 x_i$, suppose $\bar{y}_0, \bar{y}_1 > 0$. Show that $\hat{\lambda}_g = \bar{y}_g$ and $e^{\hat{\beta}_1} = \bar{y}_1/\bar{y}_0$.

Solution [Supplement].

(a) Put $p_0 = \operatorname{expit}(\beta_0)$ and $p_1 = \operatorname{expit}(\beta_0 + \beta_1)$; this is a bijection from $\mathbb{R}^2$ onto $(0,1)^2$. The log-likelihood splits into two parts with separate parameters,

$$\ell = \sum_{g=0,1} n_g \{\bar{y}_g \log p_g + (1 - \bar{y}_g) \log(1 - p_g)\}.$$

Each part is strictly concave in p_g with derivative $n_g\{\bar{y}_g/p_g - (1 - \bar{y}_g)/(1 - p_g)\}$, which vanishes only at $p_g = \bar{y}_g \in (0,1)$. So $\hat{p}_g = \bar{y}_g$, and

$$e^{\hat{\beta}_1} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1}}{e^{\hat{\beta}_0}} = \frac{\hat{p}_1/(1 - \hat{p}_1)}{\hat{p}_0/(1 - \hat{p}_0)} = \frac{\bar{y}_1/(1 - \bar{y}_1)}{\bar{y}_0/(1 - \bar{y}_0)}.$$

If some $\bar{y}_g$ is 0 or 1, the supremum is approached only as $p_g \to \bar{y}_g$, i.e. $\hat{\beta} \to \pm\infty$: this is the separation problem of Lecture 9.

(b) With $\lambda_0 = e^{\beta_0}$ and $\lambda_1 = e^{\beta_0 + \beta_1}$ (a bijection onto $(0,\infty)^2$), the log-likelihood is, up to a constant, $\sum_g n_g(\bar{y}_g \log \lambda_g - \lambda_g)$. Each term is strictly concave with derivative $n_g(\bar{y}_g/\lambda_g - 1)$, zero at $\lambda_g = \bar{y}_g$. Hence $\hat{\lambda}_g = \bar{y}_g$ and $e^{\hat{\beta}_1} = \hat{\lambda}_1/\hat{\lambda}_0 = \bar{y}_1/\bar{y}_0$. For `bioChemists` this is why $e^{0.385} = 1.47$ and $e^{0.385 + 0.247} = 1.88$ are the mean article counts of women and men. ■

Exercise 6.2 (moments of the zero-inflated Poisson)

Let $Y = (1 - Z)W$ with $Z \sim \text{Bernoulli}(\theta)$ and $W \sim \text{Poisson}(\lambda)$ independent. Show that

$$P(Y = 0) = \theta + (1 - \theta)e^{-\lambda}, \qquad \mathbb{E}Y = (1 - \theta)\lambda, \qquad \operatorname{Var} Y = (1 - \theta)\lambda(1 + \theta\lambda),$$

and conclude that a ZIP model is overdispersed whenever $\theta > 0$. Why does this mean that overdispersion alone cannot decide between a negative binomial and a ZIP model?

Solution [Supplement].

Step 1. $Y = 0$ if $Z = 1$, or if $Z = 0$ and $W = 0$; these events are disjoint, so $P(Y = 0) = \theta + (1 - \theta)e^{-\lambda}$.

Step 2. Since $(1 - Z)^2 = 1 - Z$ and Z, W are independent, $\mathbb{E}Y = (1 - \theta)\mathbb{E}W = (1 - \theta)\lambda$ and $\mathbb{E}Y^2 = (1 - \theta)\mathbb{E}W^2 = (1 - \theta)(\lambda + \lambda^2)$.

Step 3. Therefore

$$\begin{aligned}\operatorname{Var} Y &= (1 - \theta)(\lambda + \lambda^2) - (1 - \theta)^2\lambda^2 = (1 - \theta)\lambda\{1 + \lambda - (1 - \theta)\lambda\} \\ &= (1 - \theta)\lambda(1 + \theta\lambda).\end{aligned}$$

Step 4. $\operatorname{Var} Y/\mathbb{E}Y = 1 + \theta\lambda > 1$ for $\theta > 0$. The negative binomial ($\operatorname{Var}/\mathbb{E} = 1 + \mu/r$) is also overdispersed, so overdispersion only rejects Poisson. The ZIP puts the extra variability in zeros: compare the observed zero share with $\theta + (1 - \theta)e^{-\lambda}$ and the negative binomial's $P(Y = 0)$. ■

Exercise 6.3 (LOOCV from a single fit)

Let X be the $n \times p$ design matrix of a least-squares fit, $A = X^{\mathsf{T}}X$ invertible, $H = XA^{-1}X^{\mathsf{T}}$ with diagonal h_{ii}, and residuals $e_i = y_i - \hat{y}_i$. Assume that deleting row i leaves a full-rank design. Prove that the fit without observation i predicts y_i with error

$$y_i - x_i^{\mathsf{T}}\hat{\beta}_{(-i)} = \frac{e_i}{1 - h_{ii}},$$

which justifies the formula for $\mathrm{CV}_{(n)}$ in the note above.

Solution [Supplement]. Write $x = x_i$, $h = h_{ii} = x^{\mathsf{T}}A^{-1}x$ and $b = X^{\mathsf{T}}y$, so $\hat{\beta} = A^{-1}b$ and $\hat{y}_i = x^{\mathsf{T}}A^{-1}b$.

Step 1. ($h < 1$.) H is symmetric and idempotent, so $h = \sum_j h_{ij}^2 \geq h^2$ and $0 \leq h \leq 1$. Deleting row i gives $X_{(-i)}^{\mathsf{T}}X_{(-i)} = A - xx^{\mathsf{T}}$, invertible by assumption; since $\det(A - xx^{\mathsf{T}}) = \det(A)(1 - x^{\mathsf{T}}A^{-1}x)$, we get $h \neq 1$, hence $h < 1$.

Step 2. (Sherman–Morrison.) Multiplying out, using $x^{\mathsf{T}}A^{-1}x = h$,

$$(A - xx^{\mathsf{T}})\Big(A^{-1} + \frac{A^{-1}xx^{\mathsf{T}}A^{-1}}{1 - h}\Big) = I + xx^{\mathsf{T}}A^{-1}\Big(\frac{1}{1-h} - 1 - \frac{h}{1-h}\Big) = I,$$

so the second factor is $(A - xx^{\mathsf{T}})^{-1}$.

Step 3. (Leave-one-out prediction.) $\hat{\beta}_{(-i)} = (A - xx^{\mathsf{T}})^{-1}(b - xy_i)$. Since $x^{\mathsf{T}}A^{-1}(b - xy_i) = \hat{y}_i - hy_i$,

$$x^{\mathsf{T}}\hat{\beta}_{(-i)} = (\hat{y}_i - hy_i) + \frac{h(\hat{y}_i - hy_i)}{1 - h} = \frac{\hat{y}_i - hy_i}{1 - h}.$$

Step 4. Hence $y_i - x^{\mathsf{T}}\hat{\beta}_{(-i)} = \{(1-h)y_i - \hat{y}_i + hy_i\}/(1-h) = e_i/(1-h)$. ∎

Lecture 7. Regularization

7.1 Beyond least squares

Motivation. With many predictors, selecting or shrinking coefficients beats plain least squares.

The linear model $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$ is often competitive; other fitting procedures can improve accuracy and interpretability:

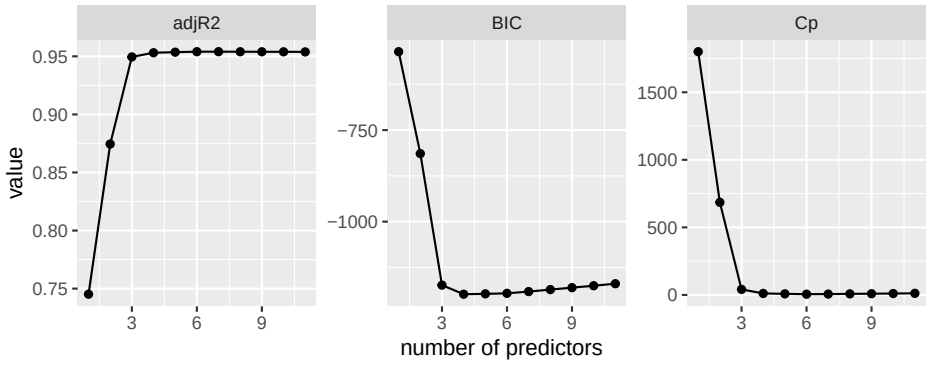
1. **subset selection**: use only some predictors;
2. **shrinkage** (regularization): shrink coefficients toward zero; some forms also select;
3. dimension reduction (not covered).

7.2 Subset selection

Fit is measured by RSS $= \sum_i (y_i - \hat{y}_i)^2$.

Best subset selection fits all 2^p models: null M_0; for each k the best M_k (smallest RSS) of the $\binom{p}{k}$ models; then chooses among $M_0, \ldots, M_p$ by an estimate of *test* error (validation, C_p, BIC, adjusted R^2), since RSS always picks the full model. `ISLR2::Credit` (balances of 400 people), `leaps::regsubsets()`:

```
fit_full <- regsubsets(Balance ~ ., data = Credit, nvmax = 11)
s <- summary(fit_full)
tibble(d = 1:11, Cp = s$cp, BIC = s$bic, adjR2 = s$adjr2) %>%
  pivot_longer(-d) %>%
  ggplot(aes(d, value)) + geom_line() + geom_point() +
  facet_wrap(~ name, scales = "free_y") + labs(x = "number of predictors")
```




```
c(Cp = which.min(s$cp), BIC = which.min(s$bic), adjR2 = which.max(s$adjr2))
```

```
##    Cp   BIC adjR2
##     6     4     7
```

BIC picks the smallest; all flatten after about four predictors.

Stepwise selection. Best subset is infeasible for large p and its search invites overfitting. **Forward stepwise** adds the best variable one at a time: $1 + \sum_{k=0}^{p-1}(p-k) = 1 + p(p+1)/2$ models (211 instead of 2^{20} for $p = 20$), not guaranteed optimal. **Backward stepwise** removes the least useful variable (needs $n > p$). **Hybrid** methods alternate adding and removing.

```
fwd <- regsubsets(Balance ~ ., data = Credit, nvmax = 11, method = "forward")
names(coef(fit_full, 4)); names(coef(fwd, 4))
```

```
## [1] "(Intercept)" "Income"      "Limit"       "Cards"       "StudentYes"
```

```
## [1] "(Intercept)" "Income"      "Limit"       "Rating"      "StudentYes"
```

Forward stepwise keeps `Rating` (entered first) instead of `Cards`.

7.2.1 Choosing the optimal model

Test error is estimated *indirectly* (adjusting training error) or *directly* (validation, CV). With d predictors and $\hat{\sigma}^2$ from the full model:

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \mathrm{BIC} = \frac{1}{n}(\mathrm{RSS} + \log(n)\, d\hat{\sigma}^2), \qquad R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}/(n-d-1)}{\mathrm{TSS}/(n-1)}.$$

- C_p penalises the optimistic training RSS by $2d\hat{\sigma}^2$; AIC (any ML model) is proportional to it for Gaussian linear models.
- BIC uses $\log n > 2$ ($n > 7$): smaller models.
- Adjusted R^2 is *maximised*, i.e. $\mathrm{RSS}/(n-d-1)$ minimised.

Why $2d\hat{\sigma}^2$ [Supplement]

For least squares with d predictors (plus intercept), $\mathbb{E}[\mathrm{RSS}] = (n-d-1)\sigma^2$, while the expected squared error on new responses at the same x's is $(n+d+1)\sigma^2$. The optimism $2(d+1)\sigma^2$ is what C_p adds back (up to the constant $2\sigma^2$; Exercise 7.3).

Cross-validation needs neither σ^2 nor a parameter count.

7.3 Regularization

Motivation. Regularization replaces yes–no variable choices by a continuous penalty.

Regularization constrains $f_\beta(x)$ against overfitting. Counting nonzero coefficients,

$$\hat{\beta} = \arg\min_\beta \frac{1}{n}\sum_i L(y_i, f_\beta(x_i)) \quad \text{subject to} \quad \sum_{j=1}^{p} \mathbb{1}(\beta_j \neq 0) \leq c,$$

is best subset selection, NP-hard with a non-convex L_0 constraint. Convex substitutes:

- $\sum_j |\beta_j| \leq c_1$ (**lasso**): sparse;
- $\sum_j \beta_j^2 \leq c_2$ (**ridge**): no zeros;
- the **elastic net**: a compromise.

The lasso combines L_2 *loss* with L_1 *penalty*.

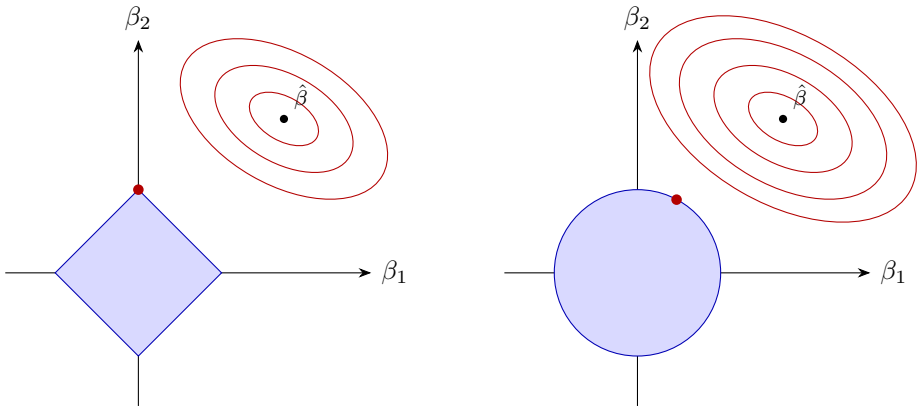


Figure. *RSS contours around $\hat{\beta}$ first touch the lasso diamond at a corner ($\beta_1 = 0$), the ridge disc at a generic point.* [Supplement]

7.3.1 Penalized form

With complexity $R(\beta)$, the constrained problem is equivalent to

$$\hat{\beta} = \arg\min_{\beta} \Big[\frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \lambda R(\beta) \Big],$$

with **tuning parameter** $\lambda \ge 0$: ridge $\lambda \sum_j \beta_j^2$, lasso $\lambda \sum_j |\beta_j|$ (intercept unpenalised, predictors standardised). The penalty **shrinks** estimates: $\lambda = 0$ is least squares, $\lambda \to \infty$ gives zero. λ is chosen by cross-validation; shrinkage trades bias for variance and can beat least squares with many predictors.

Proposition 7.1 (orthonormal case)

Let $n = p$, $X = I$ and no intercept, so least squares minimises $\sum_j (y_j - \beta_j)^2$ with solution $\hat{\beta}_j = y_j$. Then ridge and lasso give

$$\hat{\beta}_j^{\text{ridge}} = \frac{y_j}{1 + \lambda}, \qquad \hat{\beta}_j^{\text{lasso}} = \begin{cases} y_j - \lambda/2, & y_j > \lambda/2, \\ y_j + \lambda/2, & y_j < -\lambda/2, \\ 0, & |y_j| \le \lambda/2. \end{cases}$$

Proof [Supplement]**.** Both objectives separate by coordinate. Ridge: $\frac{d}{d\beta}\{(y - \beta)^2 + \lambda\beta^2\} = -2(y - \beta) + 2\lambda\beta = 0$ gives $\beta = y/(1 + \lambda)$. Lasso: $h(\beta) = (y - \beta)^2 + \lambda|\beta|$ is convex. For $\beta > 0$, $h'(\beta) = -2(y - \beta) + \lambda$ vanishes at $\beta = y - \lambda/2$, admissible iff $y > \lambda/2$; symmetrically for $\beta < 0$. If $|y| \le \lambda/2$, the one-sided derivatives at 0 are $-2y + \lambda \ge 0$ (right) and $-2y - \lambda \le 0$ (left), so 0 is the minimum. ■

Ridge shrinks proportionally; the lasso shifts by $\lambda/2$ and zeroes small ones (**soft thresholding**), hence selection.

7.4 Example: delays of flights to California

Arrival delays from EWR/JFK to four California airports on hour, origin, destination, carrier, month, weekday (slides: `tidymodels`; here `lm()`, `glmnet()`).

```
California <- flights %>%
  filter(dest %in% c("LAX", "SFO", "OAK", "SJC"), !is.na(arr_delay)) %>%
  mutate(day = as.Date(time_hour),
         dow = as.character(lubridate::wday(day, label = TRUE)),
         month = as.factor(month), hour = as.factor(hour))
dim(California)
```

```
## [1] 29836    20
```

```
set.seed(386)
idx <- sample(nrow(California), size = round(0.7 * nrow(California)))
train <- California[idx, ]; test <- California[-idx, ]
flight_model <- arr_delay ~ origin + dest + hour + carrier + month + dow
mod_reg <- lm(flight_model, data = train)
tidy(mod_reg) %>% head(4)
```

```
## # A tibble: 4 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   -16.0      6.12      -2.61 0.00905
## 2 originJFK       2.88     0.783      3.67 0.000239
## 3 destOAK        -4.61     3.10      -1.49 0.136
## 4 destSFO         1.89     0.620      3.05 0.00227
```

```
rmse <- function(y, yhat) sqrt(mean((y - yhat)^2))
rmse(test$arr_delay, predict(mod_reg, newdata = test))
```

```
## [1] 42.60868
```

`glmnet()` takes a design matrix; `alpha = 1` is the lasso, `0` ridge, values between the elastic net (`mixture` in tidymodels):

```
X_train <- model.matrix(flight_model, data = train)[, -1]
X_test <- model.matrix(flight_model, data = test)[, -1]
mod_lasso <- glmnet(X_train, train$arr_delay, alpha = 1)
coef(mod_lasso, s = 0.01)[1:4, ]
```

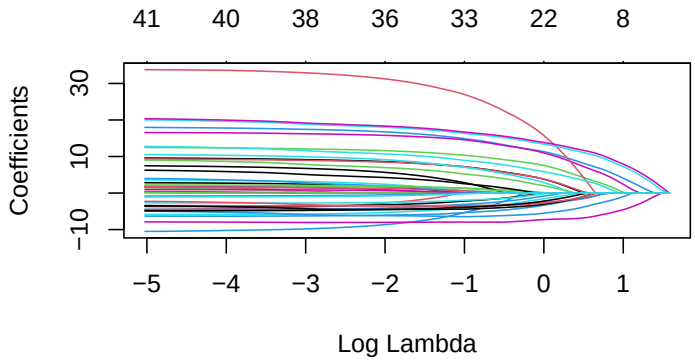
```
## (Intercept)   originJFK     destOAK     destSFO
##  -11.406671    2.783746   -4.399701    1.879023
```

```
rmse(test$arr_delay, predict(mod_lasso, newx = X_test, s = 0.01))
```

```
## [1] 42.60797
```

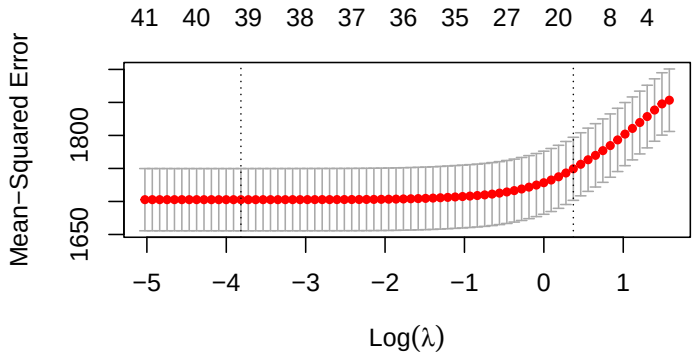
A small penalty barely moves the coefficients or test error: with 30 000 flights and 40 predictors least squares is not overfit. Coefficient paths:

```
plot(mod_lasso, xvar = "lambda")
```



10-fold CV (`cv.glmnet()`); the one-SE rule is sparser:

```
set.seed(1)
cv_lasso <- cv.glmnet(X_train, train$arr_delay, alpha = 1)
plot(cv_lasso)
```




```
c(lambda.min = cv_lasso$lambda.min, lambda.1se = cv_lasso$lambda.1se)
```

```
## lambda.min lambda.1se
##  0.0220299  1.4494202
```

```
sum(coef(cv_lasso, s = "lambda.1se") != 0)
```

```
## [1] 19
```

```
rmse(test$arr_delay, predict(cv_lasso, newx = X_test, s = "lambda.1se"))
```

```
## [1] 43.34445
```

Ridge keeps all predictors, shrunk:

```
mod_ridge <- glmnet(X_train, train$arr_delay, alpha = 0)
coef(mod_ridge, s = 0.1)[1:4, ]
```

```
## (Intercept)   originJFK     destOAK     destSFO
##   -7.683246    2.725897   -4.448581    1.861059
```

7.5 Selected exercises

Motivation. What survives of Proposition 7.1 for a general design, and where the C_p penalty comes from.

In Exercises 7.1 and 7.2, X is the $n \times p$ design matrix of centred predictors, y is centred (so no intercept is needed), and the objectives are written without the factor $1/n$, as in Proposition 7.1.

Exercise 7.1 (ridge in general)

Let $\hat\beta_\lambda$ minimise $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ with $\lambda > 0$.
(a) Show that $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ exists and is unique for every X, even when $p > n$.
(b) Show that $\|\hat\beta_\lambda\|$ is non-increasing in λ and tends to 0 as $\lambda \to \infty$.

Solution [Supplement].

(a) For $v \neq 0$, $v^\top (X^\top X + \lambda I) v = \|Xv\|^2 + \lambda\|v\|^2 > 0$, so $M = X^\top X + \lambda I$ is positive definite and invertible, whatever the rank of X. The objective equals $\beta^\top M \beta - 2\beta^\top X^\top y + \|y\|^2$, and completing

the square,

$$\beta^{\mathsf{T}} M\beta - 2\beta^{\mathsf{T}} X^{\mathsf{T}} y = (\beta - M^{-1}X^{\mathsf{T}}y)^{\mathsf{T}} M(\beta - M^{-1}X^{\mathsf{T}}y) - y^{\mathsf{T}} X M^{-1} X^{\mathsf{T}} y.$$

The first term is ≥ 0 with equality only at $\beta = M^{-1}X^{\mathsf{T}}y$, which is therefore the unique minimiser.

(b) Diagonalise $X^{\mathsf{T}}X = V\,\mathrm{diag}(d_1^2, \ldots, d_p^2)V^{\mathsf{T}}$ with V orthogonal and $d_j \geq 0$. Then $M^{-1} = V\,\mathrm{diag}\{1/(d_j^2+\lambda)\}V^{\mathsf{T}}$. With $z = V^{\mathsf{T}}X^{\mathsf{T}}y$, orthogonality of V gives

$$\|\hat\beta_\lambda\|^2 = \sum_{j=1}^{p} \frac{z_j^2}{(d_j^2+\lambda)^2},$$

each term non-increasing in λ and tending to 0. For $d_j = 1$ this is $1/(1+\lambda)$ (Proposition 7.1); small-d_j directions shrink most. ■

Exercise 7.2 (when the lasso selects nothing)

Let x_j be the jth column of X and $f(\beta) = \|y - X\beta\|^2 + \lambda\sum_j|\beta_j|$. Show that $\hat\beta = 0$ minimises f if and only if $\lambda \geq \lambda_{\max} = 2\max_j|x_j^{\mathsf{T}}y|$.

Solution [Supplement].

Step 1. (Sufficiency.) For any β,

$$\begin{aligned} f(\beta) - f(0) &= \|X\beta\|^2 - 2\sum_j \beta_j\, x_j^{\mathsf{T}} y + \lambda\sum_j|\beta_j| \\ &\geq \sum_j |\beta_j|(\lambda - 2|x_j^{\mathsf{T}}y|) \geq 0 \end{aligned}$$

when $\lambda \geq \lambda_{\max}$, using $\|X\beta\|^2 \geq 0$ and $\beta_j x_j^{\mathsf{T}}y \leq |\beta_j||x_j^{\mathsf{T}}y|$.

Step 2. (Necessity.) If $\lambda < 2|x_k^{\mathsf{T}}y|$ for some k, let $s = \mathrm{sgn}(x_k^{\mathsf{T}}y)$ and $\beta = ts\,e_k$ with $t > 0$. Then

$$f(\beta) - f(0) = t^2\|x_k\|^2 - t(2|x_k^{\mathsf{T}}y| - \lambda) < 0$$

for small t, so 0 is not a minimiser.

Step 3. With $X = I$ this is $|y_j| \leq \lambda/2$ (Proposition 7.1); `glmnet()` starts its path at $\lambda_{\max}$. ■

Exercise 7.3 (the optimism of the training error)

Let X have $q = d+1$ linearly independent columns (intercept included), $y = \mu + \epsilon$ with $\mu = X\beta$ and ϵ having uncorrelated components of mean 0 and variance σ^2, and let $y' = \mu + \epsilon'$ be new responses at the same x's, with ϵ' independent of ϵ and distributed like it. With $\hat y = Hy$, $H = X(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}$, show that

$$\mathbb{E}\,\mathrm{RSS} = \mathbb{E}\|y - \hat y\|^2 = (n-q)\sigma^2, \qquad \mathbb{E}\|y' - \hat y\|^2 = (n+q)\sigma^2.$$

Solution [Supplement].

Step 1. H is symmetric and idempotent with $HX = X$, so $H\mu = \mu$; and $\mathrm{tr}H = q$, because $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ gives $\mathrm{tr}H = \mathrm{tr}\{(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}X\} = \mathrm{tr}I_q$. For a fixed matrix B, $\mathbb{E}\,\epsilon^{\mathsf{T}}B\epsilon = \sum_{i,j} B_{ij}\mathbb{E}\epsilon_i\epsilon_j = \sigma^2\mathrm{tr}B$.

Step 2. $y - \hat y = (I-H)(\mu+\epsilon) = (I-H)\epsilon$, so $\mathrm{RSS} = \epsilon^{\mathsf{T}}(I-H)^{\mathsf{T}}(I-H)\epsilon = \epsilon^{\mathsf{T}}(I-H)\epsilon$ and $\mathbb{E}\,\mathrm{RSS} = \sigma^2\mathrm{tr}(I-H) = (n-q)\sigma^2$.

Step 3. $y' - \hat{y} = \epsilon' - H\epsilon$. Expanding, $\|y' - \hat{y}\|^2 = \|\epsilon'\|^2 - 2\epsilon'^{\mathsf{T}} H\epsilon + \epsilon^{\mathsf{T}} H\epsilon$. The cross term has mean 0 by independence and $\mathbb{E}\epsilon' = 0$, so the expectation is $n\sigma^2 + \sigma^2 \mathrm{tr} H = (n+q)\sigma^2$.

Step 4. The difference is $2q\sigma^2 = 2(d+1)\sigma^2$; C_p's $2d\hat{\sigma}^2$ corrects it up to the model-independent $2\sigma^2$. ■

Lecture 8. Classification I

8.1 Supervised learning

Motivation. Classification (phishing? churn? click?) is prediction with a categorical outcome.

Supervised learning trains on data with known outcomes: prediction (numeric) and **classification** (categorical). Clustering (no response) is **unsupervised**.

Prediction. LOESS of `volume` on `avgtemp`: the response "supervises" the fit.

```
RailTrail %>%
  mutate(loess.fit = loess(volume ~ avgtemp, data = .)$fitted) %>%
  select(volume, loess.fit, avgtemp) %>% head(4)
```

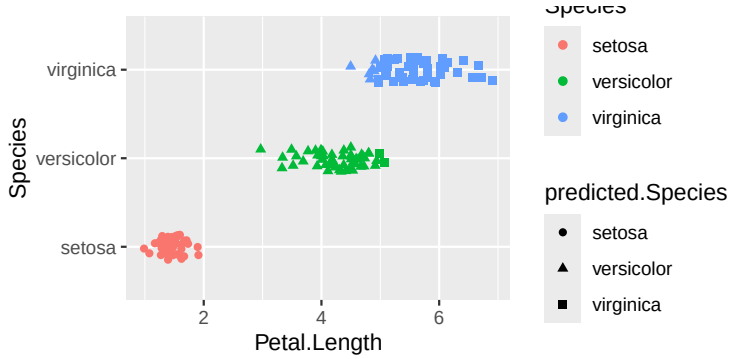
```
##   volume loess.fit avgtemp
## 1    501  469.4541    66.5
## 2    419  443.2781    61.0
## 3    397  462.1633    63.0
## 4    385  400.6197    78.0
```

Classification. Petal length almost separates the `iris` species; a crude rule (setosa < 2, versicolor 2–5, virginica > 5):

```
iris.crude <- iris %>%
  mutate(predicted.Species = ifelse(Petal.Length < 2, "setosa",
                                    ifelse(Petal.Length < 5, "versicolor", "virginica")))
iris.crude %>% count(Species, predicted.Species)
```

```
##      Species predicted.Species  n
## 1     setosa            setosa 50
## 2 versicolor        versicolor 48
## 3 versicolor         virginica  2
## 4  virginica        versicolor  6
## 5  virginica         virginica 44
```

```
ggplot(iris.crude, aes(Petal.Length, Species, colour = Species, shape = predicted.
    Species)) +
  geom_jitter(height = 0.15)
```



Two variables can be eyeballed; four or more need models or algorithms.

8.2 Naive Bayes

Motivation. Bayes' theorem turns easy $P(x \mid y)$ into the wanted $P(y \mid x)$.

Example. 1994 US Census (32 561 adults; 24% earn over \$50 000). With $y = 1$ for high earners and $x = 1$ for male, $P(y = 1) = 0.241$, $P(\text{male}) = 0.670$ and $P(\text{male} \mid y = 1) = 0.845$, so by Bayes' theorem

$$P(y = 1 \mid \text{male}) = \frac{P(\text{male} \mid y = 1)\, P(y = 1)}{P(\text{male})} = \frac{0.845 \times 0.241}{0.670} \approx 0.30.$$

Assign x^* to $y = 1$ when $P(y = 1 \mid x^*) > P(y = 0 \mid x^*)$. Naive Bayes models $P(x_1, \ldots, x_p \mid y)$ assuming **conditional independence** given the class:

$$P(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} P(x_j \mid y),$$

so for classes $k = 1, \ldots, K$,

$$P(y = i \mid x_1, \ldots, x_p) = \frac{\prod_j P(x_j \mid y = i)\; P(y = i)}{\sum_{k=1}^{K} \prod_j P(x_j \mid y = k)\; P(y = k)}.$$

Numeric predictors usually get a normal $P(x_j \mid y)$. Naive Bayes often works well. Data: 10 000 credit-card customers in `ISLR2::Default` (did they default?), split 80/20:

```
set.seed(364)
idx <- sample(nrow(Default), 0.8 * nrow(Default))
train <- Default[idx, ]; test <- Default[-idx, ]
mean(train$default == "Yes")
```

```
## [1] 0.03125
```

```
accuracy <- function(pred, truth) mean(pred == truth)
mod_nb <- naiveBayes(default ~ student + balance + income, data = train)
accuracy(predict(mod_nb, test), test$default)
```

```
## [1] 0.9625
```

Only 3% default, so always "No" is 97% accurate: accuracy is poor for unbalanced classes (Lecture 9) [Supplement].

8.3 Discriminant analysis

Motivation. With normal classes and a common covariance, the Bayes rule is linear: LDA (Fisher, 1936).

One predictor. $f_k(x) = \phi(x; \mu_k, \sigma^2)$, prior $\pi_k = P(y = k)$. With $\hat\pi_k = n_k/n$, $\hat\mu_k$, $\hat\sigma^2$, assign x to the largest **discriminant function**

$$\delta_k(x) = x\,\frac{\hat\mu_k}{\hat\sigma^2} - \frac{\hat\mu_k^2}{2\hat\sigma^2} + \log \hat\pi_k.$$

Proposition 8.1 (LDA discriminant)

Under the model above, $P(y = k \mid x)$ is largest for the k maximising $\delta_k(x)$. With two classes and $\pi_0 = \pi_1$, the boundary is $x = (\mu_0 + \mu_1)/2$.

Proof [Supplement]. $P(y = k \mid x) \propto \pi_k f_k(x)$, with the same denominator for all k. Taking logs, $\log \pi_k - \frac{(x-\mu_k)^2}{2\sigma^2} + \text{const} = \log \pi_k + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} - \frac{x^2}{2\sigma^2} + \text{const}$; the last two terms are common to all classes. With equal priors, $\delta_1(x) > \delta_0(x)$ iff $x(\mu_1 - \mu_0)/\sigma^2 > (\mu_1^2 - \mu_0^2)/(2\sigma^2)$, i.e. (for $\mu_1 > \mu_0$) iff $x > (\mu_0 + \mu_1)/2$. ∎

Several predictors. Classes $N(\mu_k, \Sigma)$:

$$f(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} \exp\Big\{ -\frac{1}{2}(x-\mu_k)^\mathsf{T}\Sigma^{-1}(x-\mu_k)\Big\}.$$

Dropping common terms of the log,

$$\delta_k(x) = x^\mathsf{T}\hat\Sigma^{-1}\hat\mu_k - \frac{1}{2}\hat\mu_k^\mathsf{T}\hat\Sigma^{-1}\hat\mu_k + \log\hat\pi_k,$$

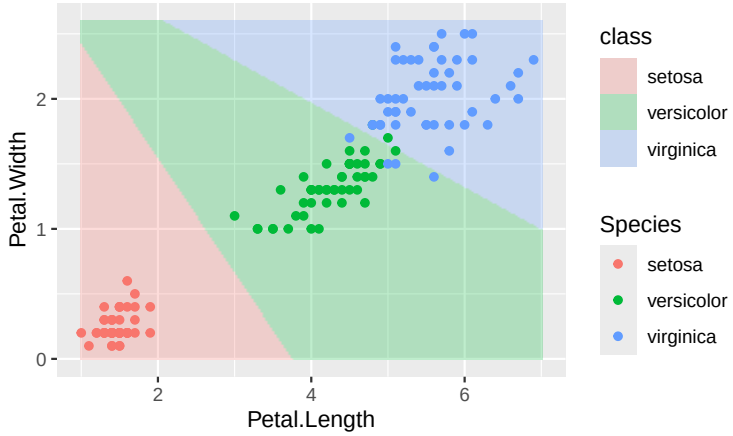
linear in x, for any number of classes.

QDA. Class-specific Σ_k: quadratic boundaries, more parameters. LDA often performs like logistic regression.

```
mod_lda <- lda(default ~ student + balance + income, data = train)
mod_qda <- qda(default ~ student + balance + income, data = train)
c(lda = accuracy(predict(mod_lda, test)$class, test$default),
  qda = accuracy(predict(mod_qda, test)$class, test$default))
```

```
##    lda    qda
## 0.9655 0.9680
```

```
lda2 <- lda(Species ~ Petal.Length + Petal.Width, data = iris)
grid <- expand_grid(Petal.Length = seq(1, 7, length = 200), Petal.Width = seq(0, 2.6,
    length = 200))
grid$class <- predict(lda2, grid)$class
ggplot(iris, aes(Petal.Length, Petal.Width)) +
  geom_raster(data = grid, aes(fill = class), alpha = 0.25) +
  geom_point(aes(colour = Species))
```



LDA boundaries on iris are straight lines.

8.4 Logistic regression for classification

Motivation. Least squares fails for logistic regression; cross-entropy, i.e. maximum likelihood, works.

$\text{logit}(q) = x^\mathsf{T}\beta$ gives $q = P(y=1 \mid x) = \text{expit}(x^\mathsf{T}\beta) = 1/(1+e^{-x^\mathsf{T}\beta})$; $q > 0.5$ iff $x^\mathsf{T}\beta > 0$, so the **decision boundary** is the hyperplane $x^\mathsf{T}\beta = 0$.

High earners. On capital gains alone the slides get $\hat\beta_0 = -1.37$, $\hat\beta_1 = 0.000335$: high-income iff gains exceed $-\hat\beta_0/\hat\beta_1 \approx \$4\,102$. On `Default` with balance:

```
mod_log_1 <- glm(default ~ balance, family = binomial, data = train)
tidy(mod_log_1)
```

```
## # A tibble: 2 x 5
##   term         estimate std.error statistic   p.value
##   <chr>           <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept) -10.8      0.421        -25.8 1.56e-146
## 2 balance       0.00559  0.000257      21.8 3.12e-105
```

```
-coef(mod_log_1)[1] / coef(mod_log_1)[2]
```

```
## (Intercept)
##     1939.69
```

Defaulter iff balance exceeds about \$1 940.

8.4.1 Why not squared loss

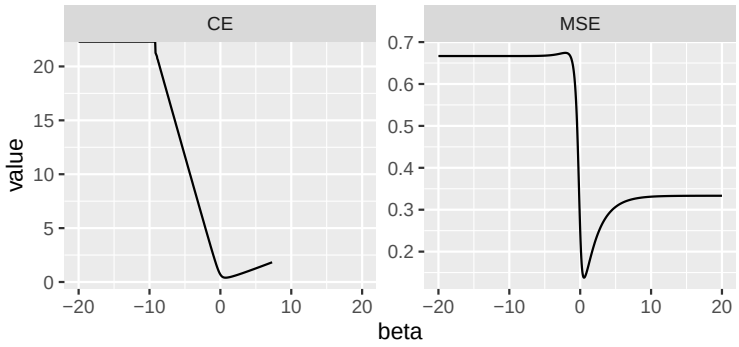
The MSE

$$\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \operatorname{expit}(x_i^\mathsf{T}\beta)\right)^2$$

is not convex in β. Toy data, $\operatorname{logit} q = \beta x$:

```
x <- c(-4, -2, -0.5, 1, 3, 5); y <- c(0, 0, 1, 0, 1, 1)
expit <- function(z) 1 / (1 + exp(-z))
tibble(beta = seq(-20, 20, by = 0.05)) %>%
  mutate(MSE = map_dbl(beta, ~ mean((y - expit(.x * x))^2)),
         CE = map_dbl(beta, ~ -mean(y * log(expit(.x * x)) + (1 - y) * log(1 - expit(.x
    * x))))) %>%
  pivot_longer(-beta) %>%
  ggplot(aes(beta, value)) + geom_line() + facet_wrap(~ name, scales = "free_y")
```

```
## Warning: Removed 254 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```
mse <- function(b) mean((y - expit(b * x))^2)
sapply(c(-20, -10.34, -2, 0.54, 5), mse)
```

```
## [1] 0.6666667 0.6666720 0.6741982 0.1378328 0.3067829
```

```
optimize(mse, c(-5, 5))$minimum
```

```
## [1] 0.5446891
```

The MSE (right) has a hump near $\beta = -2$ and a plateau at 2/3 for negative β: gradient descent from the left stalls (slides: $\beta = -10.34$), far from the minimum $\beta \approx 0.54$.

8.4.2 Cross-entropy loss

For binary y and predicted probability q, the **cross-entropy** (log) loss is

$$-\{y \log q + (1-y)\log(1-q)\}.$$

($-\log q$ if $y = 1$: infinite as $q \to 0$). It punishes confident errors and its average

$$-\frac{1}{n}\sum_{i=1}^{n}\left\{y_i \log \operatorname{expit}(x_i^\top \beta) + (1-y_i)\log\left(1-\operatorname{expit}(x_i^\top \beta)\right)\right\},$$

is convex in β (left panel).

Convexity of cross-entropy [Supplement]

With $q_i = \operatorname{expit}(x_i^\top \beta)$, the gradient of the average loss is $\frac{1}{n}\sum_i (q_i - y_i)x_i$ and the Hessian is $\frac{1}{n}\sum_i q_i(1-q_i)x_i x_i^\top$, positive semidefinite; Newton's method (IRLS, as in `glm()`) finds the global minimum when it exists.

Connection to maximum likelihood. Given x_i, $y_i \sim \text{Bernoulli}(q_i)$ with likelihood $q_i^{y_i}(1-q_i)^{1-y_i}$, so the log-likelihood is

$$\sum_{i=1}^{n}\{y_i \log q_i + (1-y_i)\log(1-q_i)\},$$

$-n$ times the average cross-entropy: minimising cross-entropy *is* maximum likelihood (as least squares is for normal errors, ridge/lasso with a prior on β).

More than two classes. Chains of binary models or the multinomial model (`nnet::multinom()`); discriminant analysis is more common.

Summary. Naive Bayes: conditional independence; LDA/QDA: normal classes; logistic regression: linear logit $P(y = 1 \mid x)$ fitted by maximum likelihood.

8.5 Selected exercises

Motivation. Naive Bayes (binary predictors) and LDA give a linear log-odds, the logistic form; QDA does not.

Exercise 8.1 (naive Bayes is linear in the log-odds)

Two classes $y \in \{0, 1\}$ with priors π_0, π_1, and binary predictors $x_1, \ldots, x_p$ that are conditionally independent given y, with $p_{jk} = P(x_j = 1 \mid y = k) \in (0, 1)$. Show that

$$\log \frac{P(y=1 \mid x)}{P(y=0 \mid x)} = w_0 + \sum_{j=1}^{p} w_j x_j,$$

and give $w_0, \ldots, w_p$. How does the naive Bayes fit of these coefficients differ from a logistic regression fit?

Solution [Supplement].

Step 1. By Bayes' theorem the denominator cancels in the ratio, and by conditional independence $P(x \mid y = k) = \prod_j p_{jk}^{x_j}(1-p_{jk})^{1-x_j}$. Hence

$$\log \frac{P(y=1 \mid x)}{P(y=0 \mid x)} = \log \frac{\pi_1}{\pi_0} + \sum_j \left\{x_j \log \frac{p_{j1}}{p_{j0}} + (1-x_j)\log \frac{1-p_{j1}}{1-p_{j0}}\right\}.$$

Step 2. Collecting the terms in x_j,

$$w_j = \log \frac{p_{j1}(1-p_{j0})}{p_{j0}(1-p_{j1})}, \qquad w_0 = \log \frac{\pi_1}{\pi_0} + \sum_j \log \frac{1-p_{j1}}{1-p_{j0}};$$

w_j is the log odds ratio between x_j and y.

Step 3. Same family of posteriors. Naive Bayes estimates each w_j from class-wise frequencies of x_j alone; logistic regression fits all w_j jointly by maximum likelihood of y given x, adjusting for correlation (naive Bayes double-counts correlated evidence). ■

Exercise 8.2 (LDA implies a logistic model)

Let $x \mid y = k \sim N(\mu_k, \Sigma)$ for $k = 0, 1$, with priors π_0, π_1 and Σ positive definite. Show that $P(y = 1 \mid x) = \operatorname{expit}(\beta_0 + \beta^\mathsf{T} x)$ with $\beta = \Sigma^{-1}(\mu_1 - \mu_0)$, and find β_0. Show that for $\pi_0 = \pi_1$ the decision boundary passes through $(\mu_0 + \mu_1)/2$.

Solution [Supplement].

Step 1. The normalising constants of the two densities are equal (same Σ), so

$$\log \frac{\pi_1 f_1(x)}{\pi_0 f_0(x)} = \log \frac{\pi_1}{\pi_0} - \frac{1}{2}(x-\mu_1)^\mathsf{T}\Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_0)^\mathsf{T}\Sigma^{-1}(x-\mu_0).$$

Step 2. The quadratic terms $x^\mathsf{T}\Sigma^{-1}x$ cancel, and by symmetry of Σ^{-1} the cross terms give $x^\mathsf{T}\Sigma^{-1}(\mu_1 - \mu_0)$. So the log-odds equals $\beta_0 + \beta^\mathsf{T}x$ with

$$\beta = \Sigma^{-1}(\mu_1 - \mu_0), \qquad \beta_0 = \log \frac{\pi_1}{\pi_0} - \frac{1}{2}(\mu_1^\mathsf{T}\Sigma^{-1}\mu_1 - \mu_0^\mathsf{T}\Sigma^{-1}\mu_0),$$

and $P(y = 1 \mid x) = \operatorname{expit}(\beta_0 + \beta^\mathsf{T}x)$.

Step 3. For $\pi_0 = \pi_1$ and $m = (\mu_0 + \mu_1)/2$, $\beta^\mathsf{T}m = \frac{1}{2}(\mu_1 - \mu_0)^\mathsf{T}\Sigma^{-1}(\mu_1 + \mu_0) = \frac{1}{2}(\mu_1^\mathsf{T}\Sigma^{-1}\mu_1 - \mu_0^\mathsf{T}\Sigma^{-1}\mu_0) = -\beta_0$, because the mixed terms $\mu_1^\mathsf{T}\Sigma^{-1}\mu_0$ and $\mu_0^\mathsf{T}\Sigma^{-1}\mu_1$ are equal. So m lies on the boundary $\beta_0 + \beta^\mathsf{T}x = 0$, which generalises Proposition 8.1.

Step 4. The converse fails (non-normal classes can give linear log-odds): logistic regression is more robust, LDA more efficient under normality. ■

Exercise 8.3 (QDA boundaries can be two points)

Let $x \mid y = k \sim N(\mu_k, \sigma_k^2)$ for $k = 0, 1$, with equal priors.
(a) Show that the set where the Bayes rule chooses class 1 is defined by a quadratic inequality in x when $\sigma_0 \neq \sigma_1$.
(b) For $\mu_0 = \mu_1 = 0$, $\sigma_0 = 1$, $\sigma_1 = 2$, find the rule. What does LDA do with these data?

Solution [Supplement].

(a) Class 1 is chosen when $\log f_1(x) > \log f_0(x)$, i.e.

$$-\log \sigma_1 - \frac{(x-\mu_1)^2}{2\sigma_1^2} > -\log \sigma_0 - \frac{(x-\mu_0)^2}{2\sigma_0^2}.$$

The coefficient of x^2 in the difference is $\frac{1}{2\sigma_0^2} - \frac{1}{2\sigma_1^2} \neq 0$, so this is a quadratic inequality; its solution set is an interval or the complement of an interval (possibly empty or all of $\mathbb{R}$), with at most two boundary points.

(b) $-\log 2 - x^2/8 > -x^2/2$, i.e. $\frac{3}{8}x^2 > \log 2$, i.e. $|x| > \sqrt{8 \log 2/3} \approx 1.36$. LDA pools variances; with $\mu_0 = \mu_1$ its discriminants differ only by $\log \hat{\pi}_k$, so it assigns every point to one class. ■

Lecture 9. Classification II

9.1 Logistic regression as a classifier

Motivation. Thresholds turn probabilities into decisions; judging them needs more than accuracy, especially for rare classes.

Logistic regression predicts $P(y = 1 \mid x) = \operatorname{expit}(x^\top \beta)$, is fitted with cross-entropy loss, and classifies $y = 1$ when $\hat{p} \geq 0.5$.

A coin example. Tosses $0, 0, 0, 1, 1, 0, 0, 0, 1, 1$: $4 \log \theta + 6 \log(1-\theta)$ has derivative $4/\theta - 6/(1-\theta) = 0$ at $\theta = 0.4$, the sample mean (minimum cross-entropy).

The null model. (Slides: census, 24% high earners.) `ISLR2::Default`, split 80/20 as in Lecture 8. $P(y = 1) = \operatorname{expit}(\beta_0)$ gives everyone $\bar{\pi} < 0.5$, so all are classified $y = 0$.

```
set.seed(364)
idx <- sample(nrow(Default), 0.8 * nrow(Default))
train <- Default[idx, ]; test <- Default[-idx, ]
train %>% count(default) %>% mutate(pct = n / sum(n))
```

```
##   default    n     pct
## 1      No 7750 0.96875
## 2     Yes  250 0.03125
```

9.1.1 Thresholding

The decision rule need not use 0.5:

$$\text{classify}(x) = \begin{cases} 1, & \hat{P}(y = 1 \mid x) \geq t, \\ 0, & \text{otherwise.} \end{cases}$$

Raising t: fewer false positives, more false negatives.

9.1.2 Accuracy

Accuracy: proportion classified correctly. Null, `balance` only, all predictors:

```
fit_null <- glm(default ~ 1, family = binomial, data = train)
fit_1 <- glm(default ~ balance, family = binomial, data = train)
fit_all <- glm(default ~ student + balance + income, family = binomial, data = train)
classify <- function(fit, data, t = 0.5)
  factor(ifelse(predict(fit, data, type = "response") >= t, "Yes", "No"), levels = c("No
    ", "Yes"))
map_dbl(list(null = fit_null, bal = fit_1, all = fit_all),
        ~ mean(classify(.x, train) == train$default))
```

```
##     null      bal      all
## 0.968750 0.974125 0.974875
```

Small gains: the null model is already 97% accurate.

9.2 Evaluating classifiers

9.2.1 Drawbacks of accuracy

Labelling 100 e-mails (2 spam) all as ham is 98% accurate and useless.

9.2.2 The confusion matrix

Call class 1 positive and class 0 negative.

	predicted 1	predicted 0
actual 1	TP (true positive)	FN (false negative: missed)
actual 0	FP (false positive: false alarm)	TN (true negative)

$$\text{accuracy} = \frac{TP + TN}{n}, \qquad \text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}.$$

Precision: share of predicted 1s that are 1; **recall**: share of actual 1s predicted 1. All-ham filter: precision $0/0$, recall $0/2 = 0$.

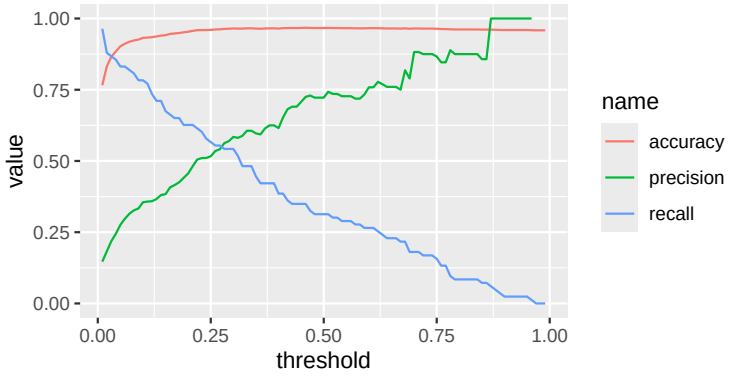
```
table(predicted = classify(fit_1, test), actual = test$default)
```

```
##          actual
## predicted   No  Yes
##       No  1907   58
##       Yes   10   25
```

The trade-off. Higher thresholds raise precision (fewer FP), lower ones recall (fewer FN; $t = 0$ gives recall 1). Accuracy peaks in between. Disease screening favours recall; a life sentence favours precision.

Other names: **sensitivity** = recall = true positive rate (TPR); **specificity** = true negative rate $= TN/(TN + FP)$; 1−specificity = false positive rate (FPR); precision = positive predictive value (PPV).

```
p_test <- predict(fit_all, test, type = "response")
y_test <- test$default == "Yes"
tibble(t = seq(0.01, 0.99, by = 0.01)) %>%
  mutate(accuracy = map_dbl(t, ~ mean((p_test >= .x) == y_test)),
         precision = map_dbl(t, ~ sum(p_test >= .x & y_test) / sum(p_test >= .x)),
         recall = map_dbl(t, ~ sum(p_test >= .x & y_test) / sum(y_test))) %>%
  pivot_longer(-t) %>%
  ggplot(aes(t, value, colour = name)) + geom_line(na.rm = TRUE) + labs(x = "threshold")
```



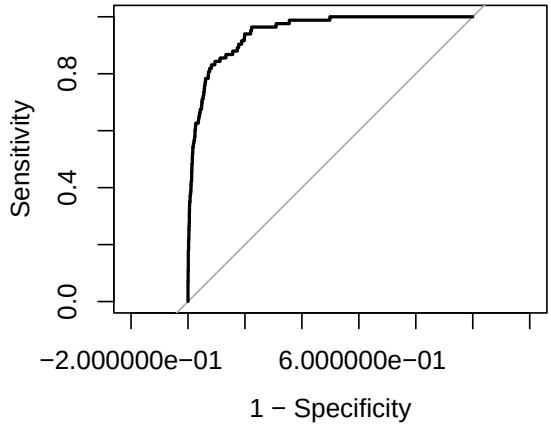
9.2.3 ROC and precision–recall curves

Over all thresholds: precision against recall is the **PR curve**; TPR against FPR is the **ROC curve**, best near the top-left. **AUC**: 1 perfect, 0.5 random. PR is more informative for unbalanced classes.

```
roc_1 <- roc(test$default, predict(fit_1, test, type = "response"), quiet = TRUE)
roc_all <- roc(test$default, p_test, quiet = TRUE)
c(auc_balance = auc(roc_1), auc_all = auc(roc_all))
```

```
## auc_balance     auc_all
##   0.9468359   0.9467919
```

```
plot(roc_all, legacy.axes = TRUE)
```



What AUC measures [Supplement]

AUC equals the probability that a randomly chosen positive case receives a higher predicted probability than a randomly chosen negative case (Mann–Whitney; Exercise 9.1): ranking quality, free of threshold and class proportions.

9.2.4 Training versus testing

Slides, census data:

model	accuracy (train)	accuracy (test)	sensitivity (test)	specificity (test)
all predictors	0.852	0.849	0.598	0.928
capital gain only	0.800	0.803	0.204	0.991
null	0.759	0.761	0.000	1.000

Test accuracy is slightly lower; the full model trades a little specificity for much sensitivity. For `Default`:

```
eval_model <- function(fit) {
  pr <- classify(fit, test)
  c(acc_train = mean(classify(fit, train) == train$default),
    acc_test = mean(pr == test$default),
    sens = mean(pr[test$default == "Yes"] == "Yes"),
    spec = mean(pr[test$default == "No"] == "No"))
}
round(rbind(all = eval_model(fit_all), balance = eval_model(fit_1), null = eval_model(
    fit_null)), 3)
```

```
##         acc_train acc_test  sens  spec
## all         0.975    0.967 0.313 0.995
## balance     0.974    0.966 0.301 0.995
## null        0.969    0.959 0.000 1.000
```

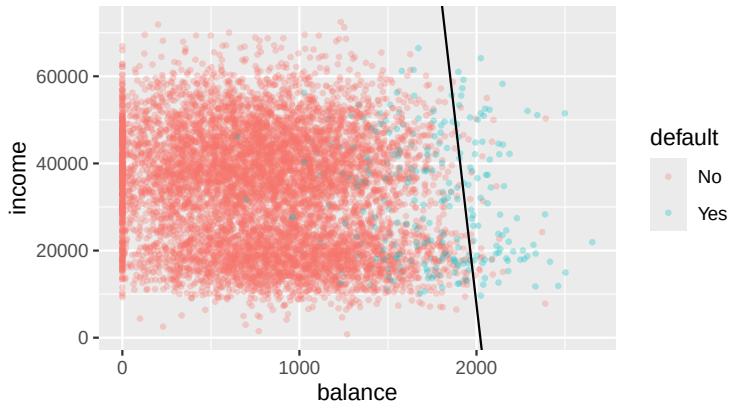
At $t = 0.5$ most defaulters are missed; lower t if misses are costly.

9.3 Linear separability and regularization

Motivation. Under perfect separation the ML coefficients diverge; regularization fixes this.

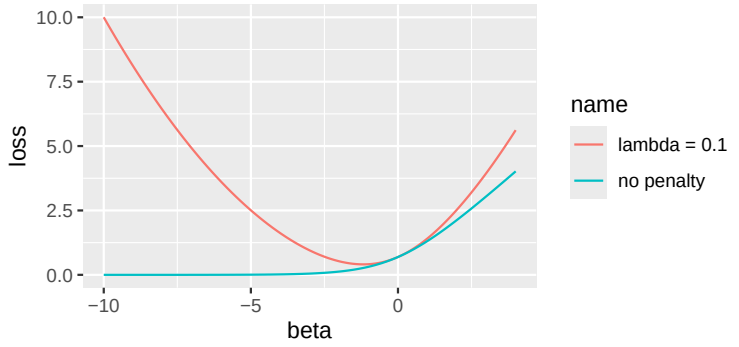
Decision boundaries. Slides: $\operatorname{logit}\hat{\pi} = -3.085 + 0.046 \cdot \text{hours}$, boundary about 67 hours; with $\hat{\pi} = \operatorname{expit}(-4.85 + 0.047 \cdot \text{hours} + 0.042 \cdot \text{age})$ it is a line in the (hours, age) plane.

```
fit_2 <- glm(default ~ balance + income, family = binomial, data = train)
b <- coef(fit_2)
ggplot(train, aes(balance, income, colour = default)) +
  geom_point(alpha = 0.3, size = 0.8) +
  geom_abline(intercept = -b[1] / b[3], slope = -b[2] / b[3])
```



Perfect separation. Points $(-1, 1)$, $(1, 0)$, model $\operatorname{expit}(\beta_1 x)$: the loss decreases to 0 as $\beta_1 \to -\infty$, predicting exactly 0 or 1; a new $(0.5, 1)$ would have *infinite* loss.

```
x <- c(-1, 1); y <- c(1, 0); expit <- function(z) 1 / (1 + exp(-z))
ce <- function(b, lambda = 0)
  -mean(y * log(expit(b * x)) + (1 - y) * log(1 - expit(b * x))) + lambda * b^2
tibble(beta = seq(-10, 4, by = 0.05)) %>%
  mutate(`no penalty` = map_dbl(beta, ce), `lambda = 0.1` = map_dbl(beta, ce, lambda =
    0.1)) %>%
  pivot_longer(-beta) %>%
  ggplot(aes(beta, value, colour = name)) + geom_line() + labs(y = "loss")
```



Linearly separable: a hyperplane separates the classes; coefficients diverge (`glm()` warns of fitted probabilities 0 or 1).

Regularized logistic regression adds a penalty, e.g. L_2:

$$-\frac{1}{n}\sum_{i=1}^{n}\left\{y_i \log \operatorname{expit}(x_i^\top \beta) + (1 - y_i)\log\left(1 - \operatorname{expit}(x_i^\top \beta)\right)\right\} + \lambda \sum_{j=1}^{p} \beta_j^2.$$

It has a finite minimum even under separation (Exercise 9.3). Standardise, then `glmnet(..., family = "binomial")`:

```
X <- scale(model.matrix(default ~ student + balance + income, data = train)[, -1])
cv_fit <- cv.glmnet(X, train$default, family = "binomial", alpha = 0)
coef(cv_fit, s = "lambda.min")
```

```
## 4 x 1 sparse Matrix of class "dgCMatrix"
##                       s1
## (Intercept) -4.83965221
## studentYes  -0.11306543
## balance      1.81973290
## income       0.01566196
```

9.4 Selected exercises

Motivation. Proofs of three claims of the lecture.

Exercise 9.1 (AUC as a ranking probability)

A classifier gives scores $s_1, \dots, s_m$ to the positive cases and $t_1, \dots, t_k$ to the negative cases, all $m + k$ scores distinct. Show that the area under the ROC curve equals

$$\mathrm{AUC} = \frac{1}{mk} \sum_{i=1}^{m} \sum_{j=1}^{k} \mathbb{1}(s_i > t_j),$$

the probability that a randomly chosen positive case scores higher than a randomly chosen negative case.

Solution [Supplement].

Step 1. At threshold c the classifier has $\mathrm{TPR}(c) = \#\{i : s_i \geq c\}/m$ and $\mathrm{FPR}(c) = \#\{j : t_j \geq c\}/k$. Lowering c moves $(\mathrm{FPR}, \mathrm{TPR})$ from $(0, 0)$ to $(1, 1)$, up $1/m$ at each positive score and right $1/k$ at each negative one, one at a time (distinct scores): a staircase.

Step 2. The area is the sum over horizontal steps of width times height: the step at t_j has width $1/k$ and height $\#\{i : s_i > t_j\}/m$.

Step 3. Summing over j gives the formula: the share of the mk (positive, negative) pairs ordered correctly. (Ties give diagonal segments, counting $\frac{1}{2}$.) ■

Exercise 9.2 (precision depends on prevalence)

A test has sensitivity se and specificity sp, and the positive class has prevalence $\pi = P(y = 1)$. Show that

$$\text{precision} = P(y = 1 \mid \hat{y} = 1) = \frac{\mathrm{se}\,\pi}{\mathrm{se}\,\pi + (1 - \mathrm{sp})(1 - \pi)}.$$

Evaluate it for $\mathrm{se} = \mathrm{sp} = 0.99$ with $\pi = 0.5$ and with $\pi = 0.001$, and explain why the ROC curve but not the precision–recall curve is unaffected by the class proportions.

Solution [Supplement].

Step 1. By definition $\mathrm{se} = P(\hat{y} = 1 \mid y = 1)$ and $1 - \mathrm{sp} = P(\hat{y} = 1 \mid y = 0)$. By the law of total probability $P(\hat{y} = 1) = \mathrm{se}\,\pi + (1 - \mathrm{sp})(1 - \pi)$, and Bayes' theorem gives the formula.

Step 2. $\pi = 0.5$: 0.99. $\pi = 0.001$: $0.00099/(0.00099 + 0.00999) = 0.090$, nine in ten positives are false alarms.

Step 3. Sensitivity and specificity condition on the true class, so the ROC curve does not depend on π; precision mixes the classes through π, so the PR curve shifts with prevalence. ■

Exercise 9.3 (separation and the ridge penalty)

Let $L(\beta)$ be the mean cross-entropy loss of logistic regression for data (x_i, y_i), where x_i includes the constant 1 and all coefficients are penalised.
(a) Suppose some b satisfies $b^{\mathsf{T}} x_i > 0$ whenever $y_i = 1$ and $b^{\mathsf{T}} x_i < 0$ whenever $y_i = 0$. Show that L has no minimiser.
(b) Show that for every $\lambda > 0$ the penalised loss $L_\lambda(\beta) = L(\beta) + \lambda\|\beta\|^2$ has exactly one minimiser, whether or not the data are separable.

Solution [Supplement].

(a) Each term of L is $\log(1+e^{-\eta_i})$ if $y_i = 1$ and $\log(1+e^{\eta_i})$ if $y_i = 0$, with $\eta_i = \beta^{\mathsf{T}} x_i$; both are strictly positive, so $L(\beta) > 0$ for every finite β. Along $\beta = tb$ every $y_i = 1$ term is $\log(1 + e^{-tb^{\mathsf{T}} x_i}) \to 0$ and every $y_i = 0$ term is $\log(1 + e^{tb^{\mathsf{T}} x_i}) \to 0$ as $t \to \infty$. Hence $\inf L = 0$, and the infimum is not attained.

(b) *Existence.* L_λ is continuous and $L_\lambda(\beta) \geq \lambda\|\beta\|^2$ because $L \geq 0$. $K = \{\beta : L_\lambda(\beta) \leq L_\lambda(0)\}$ is non-empty, closed and inside $\|\beta\|^2 \leq L_\lambda(0)/\lambda$, hence compact; the minimum on K is global since $L_\lambda > L_\lambda(0)$ outside K.

Uniqueness. L is convex (its Hessian $\frac{1}{n}\sum_i q_i(1-q_i)x_i x_i^{\mathsf{T}}$ is positive semidefinite, Lecture 8) and $\lambda\|\beta\|^2$ is strictly convex, so L_λ is strictly convex. If $\beta \neq \beta'$ were both minimisers with value v, then $L_\lambda(\frac{\beta+\beta'}{2}) < \frac{1}{2}L_\lambda(\beta) + \frac{1}{2}L_\lambda(\beta') = v$, a contradiction. ■

Lecture 10. Splines and Generalized Additive Models

10.1 Beyond linearity

Motivation. Relax linearity step by step while keeping interpretability.

Polynomials and step functions, regression and smoothing splines, local regression, GAMs. Example: `ISLR2::Wage`, 3000 male workers in the US Mid-Atlantic.

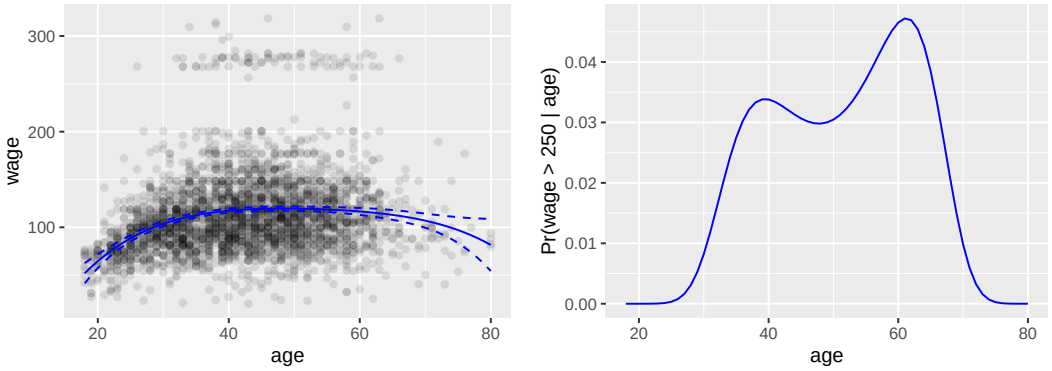
10.2 Polynomial regression and step functions

Polynomial regression replaces the line by

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \epsilon_i,$$

linear in the coefficients (least squares). Degrees above 4 behave strangely near the boundary. We look at $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \cdots + \hat{\beta}_4 x_0^4$ with a ± 2 SE band, not the coefficients.

```
fit4 <- lm(wage ~ poly(age, 4), data = Wage)
grid <- tibble(age = 18:80)
pr <- predict(fit4, grid, se.fit = TRUE)
grid <- grid %>% mutate(fit = pr$fit, lo = fit - 2 * pr$se.fit, hi = fit + 2 * pr$se.fit
    )
ggplot(Wage, aes(age, wage)) + geom_point(alpha = 0.1) +
  geom_line(data = grid, aes(y = fit), colour = "blue") +
  geom_line(data = grid, aes(y = lo), colour = "blue", linetype = "dashed") +
  geom_line(data = grid, aes(y = hi), colour = "blue", linetype = "dashed")
fit_log <- glm(I(wage > 250) ~ poly(age, 4), family = binomial, data = Wage)
grid %>% mutate(p = predict(fit_log, grid, type = "response")) %>%
  ggplot(aes(age, p)) + geom_line(colour = "blue") + labs(y = "Pr(wage > 250 | age)")
```



A high-earner group (above \$250 000) stands apart; logistic regression for $\mathbb{1}(\text{wage} > 250)$:

$$P(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)};$$

its intervals (not drawn) are wide: few high earners.

Step functions avoid global structure: cut X at $c_1 < \cdots < c_K$ into $K+1$ bins,

$$C_0(X) = I(X < c_1),\ C_1(X) = I(c_1 \le X < c_2),\ \ldots,\ C_K(X) = I(c_K \le X),$$

summing to 1. In $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \epsilon_i$, β_0 is the mean for $X < c_1$ and $\beta_0 + \beta_j$ the mean in $[c_j, c_{j+1})$.

```
coef(lm(wage ~ cut(age, 4), data = Wage))
```

```
##           (Intercept)   cut(age, 4)(33.5,49]   cut(age, 4)(49,64.5]
##             94.158392              24.053491              23.664559
## cut(age, 4)(64.5,80.1]
##              7.640592
```

They miss within-bin trends.

10.3 Regression splines

Motivation. Splines are local like step functions and smooth like polynomials.

Basis functions. Polynomial and step regression are special cases of

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \epsilon_i$$

with fixed basis functions ($b_j(x) = x^j$; $b_j(x) = I(c_j \le x < c_{j+1})$): a linear model in $b_j(x_i)$.

Piecewise polynomials. Coefficients change at **knots**. One knot at c:

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \epsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \epsilon_i, & x_i \ge c. \end{cases}$$

Requiring continuity and continuous first and second derivatives at c gives a **cubic spline**: the right piece keeps 1 of its 4 degrees of freedom. K knots: $K + 4$ degrees of freedom.

Spline basis. $y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \epsilon_i$: x, x^2, x^3 plus one **truncated power function** per knot,

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x - \xi)^3, & x > \xi, \\ 0, & \text{otherwise.} \end{cases}$$

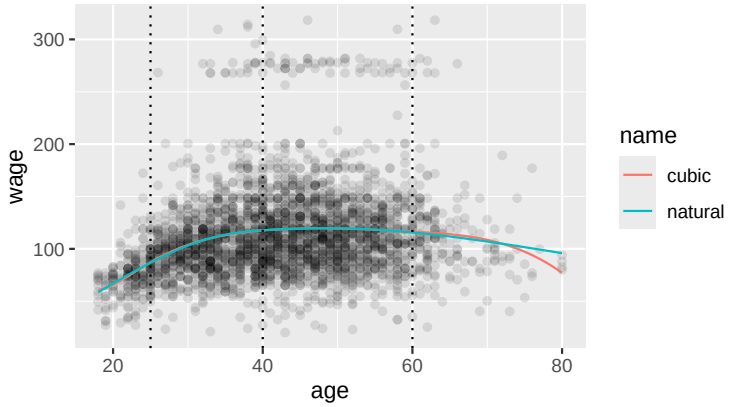
which breaks only the third derivative at ξ: $K + 4$ coefficients.

Why the truncated power basis works [Supplement]

$h(x, \xi)$ and its first two derivatives, $3(x - \xi)_+^2$ and $6(x - \xi)_+$, vanish at $x = \xi$ from both sides, while the third derivative jumps from 0 to 6. Conversely, any cubic spline with knots $\xi_1 < \cdots < \xi_K$ differs on $[\xi_k, \xi_{k+1})$ from its expression on $[\xi_{k-1}, \xi_k)$ by a cubic that vanishes with two derivatives at ξ_k, which must be $c\,(x - \xi_k)^3$. So the $K + 4$ functions span exactly the cubic splines.

`bs()` gives a stabler basis (B-splines) of the same space:

```
fit_bs <- lm(wage ~ bs(age, knots = c(25, 40, 60)), data = Wage)
fit_ns <- lm(wage ~ ns(age, df = 4), data = Wage)
grid <- tibble(age = 18:80) %>%
  mutate(cubic = predict(fit_bs, .), natural = predict(fit_ns, .))
ggplot(Wage, aes(age, wage)) + geom_point(alpha = 0.1) +
  geom_line(data = pivot_longer(grid, -age), aes(y = value, colour = name)) +
  geom_vline(xintercept = c(25, 40, 60), linetype = "dotted")
```



Natural splines (`ns()`) are linear beyond the boundary knots, taming the high variance at the ends.

Knots. Usually at quantiles of X (`ns(age, df = 4)`); their number by cross-validation:

```
set.seed(1)
folds <- sample(rep(1:10, length.out = nrow(Wage)))
cv_ns <- function(df) mean(map_dbl(1:10, function(k) {
  fit <- lm(wage ~ ns(age, df = df), data = Wage[folds != k, ])
  mean((Wage$wage[folds == k] - predict(fit, Wage[folds == k, ]))^2)
}))
round(map_dbl(1:8, cv_ns), 1)
```

```
## [1] 1676.8 1602.4 1595.4 1595.2 1595.1 1594.0 1593.9 1593.6
```

CV error drops up to about 3 df, then is flat. Natural splines beat equal-df polynomials at the boundaries (knots, not high powers).

10.4 Smoothing splines

Motivation. A knot at every point, with a roughness penalty instead of knot choice.

An unconstrained g interpolates (RSS $= 0$). The **smoothing spline** minimises

$$\sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int g''(t)^2 \, dt, \qquad \lambda \geq 0.$$

"Loss + penalty": $\int g''^2$ measures roughness. $\lambda = 0$ interpolates; $\lambda \to \infty$ gives a line. The minimiser is a shrunken natural cubic spline with knots at the unique x_i.

Effective degrees of freedom. With $\hat{g}_\lambda = S_\lambda y$ ($n \times n$ smoother matrix),

$$\mathrm{df}_\lambda = \sum_{i=1}^{n} \{S_\lambda\}_{ii},$$

decreasing from n to 2 (Exercise 10.3); for least squares it is the number of parameters [Supplement].

Choosing λ. LOOCV from one fit:

$$\mathrm{RSS}_{\mathrm{cv}}(\lambda) = \sum_{i=1}^{n} \big(y_i - \hat{g}_\lambda^{(-i)}(x_i)\big)^2 = \sum_{i=1}^{n} \left[\frac{y_i - \hat{g}_\lambda(x_i)}{1 - \{S_\lambda\}_{ii}}\right]^2,$$

with $\hat{g}_\lambda^{(-i)}$ fitted without observation i.

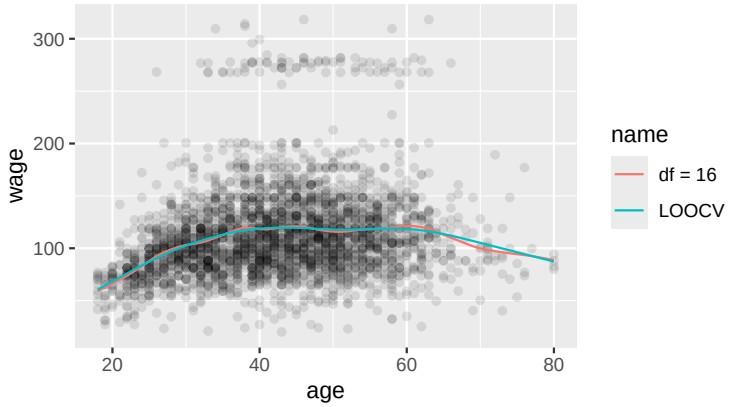
```
ss16 <- smooth.spline(Wage$age, Wage$wage, df = 16)
ss_cv <- smooth.spline(Wage$age, Wage$wage, cv = TRUE)
```

```
## Warning in smooth.spline(Wage$age, Wage$wage, cv = TRUE): cross-validation with non-unique 'x'
##     values seems doubtful
```

```
c(df_chosen = ss_cv$df, lambda = ss_cv$lambda)
```

```
##  df_chosen     lambda
## 6.79459570 0.02792303
```

```
tibble(age = 18:80, `df = 16` = predict(ss16, 18:80)$y,
       `LOOCV` = predict(ss_cv, 18:80)$y) %>%
  pivot_longer(-age) %>%
  ggplot(aes(age, value, colour = name)) +
  geom_point(data = Wage, aes(age, wage), inherit.aes = FALSE, alpha = 0.1) +
  geom_line() + labs(y = "wage")
```

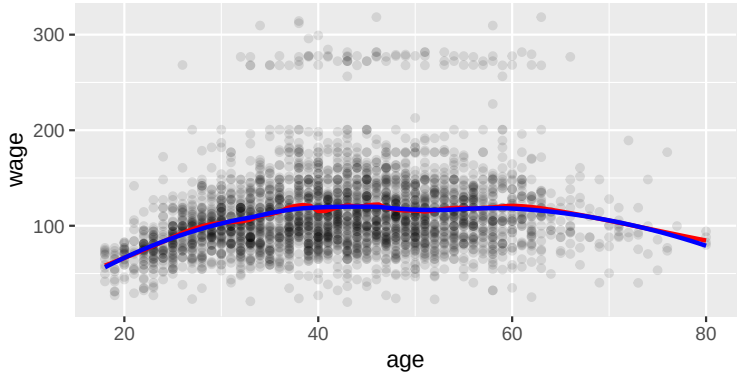


LOOCV (about 7 df) is smoother than 16 df and fits almost as well.

10.5 Local regression

LOESS (Lecture 2) fits at each x_0 a weighted line on the nearest fraction s (**span**) of points; smaller span, wigglier fit. It is memory-based and poor beyond three or four dimensions.

```
ggplot(Wage, aes(age, wage)) + geom_point(alpha = 0.1) +
  geom_smooth(method = "loess", span = 0.2, se = FALSE, colour = "red", formula = y ~ x)
    +
  geom_smooth(method = "loess", span = 0.5, se = FALSE, colour = "blue", formula = y ~ x
    )
```



10.6 Generalized additive models

Motivation. GAMs add one smooth function per predictor.

A GAM replaces each $\beta_j x_{ij}$ by a smooth function:

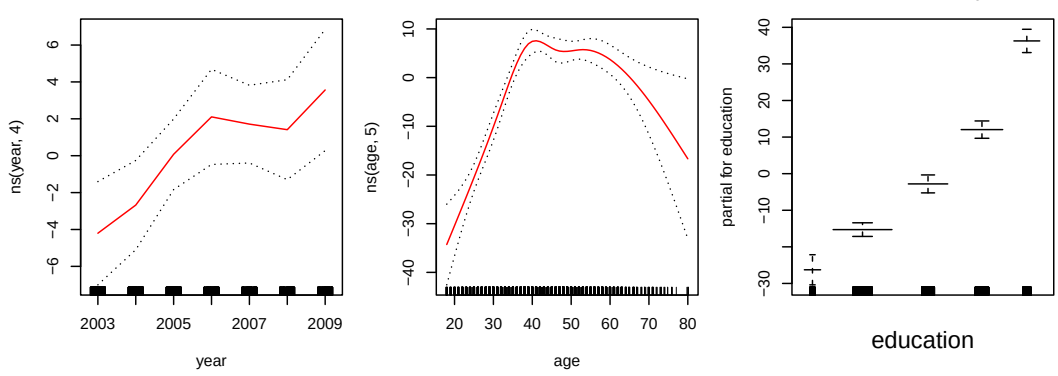
$$y_i = \beta_0 + \sum_{j=1}^{p} f_j(x_{ij}) + \epsilon_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \epsilon_i.$$

Additive: separate f_j per X_j, built from the smoothers above. For the wage data,

$$\text{wage} = \beta_0 + f_1(\text{year}) + f_2(\text{age}) + f_3(\text{education}) + \epsilon,$$

with natural splines for year and age and dummies for education: one big regression, fitted by `lm()`:

```
gam1 <- lm(wage ~ ns(year, 4) + ns(age, 5) + education, data = Wage)
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot.Gam(gam1, se = TRUE, col = "red")
```



Others fixed, wage rises slightly with year, peaks in middle age, rises with education. Smoothing splines: `gam::gam(wage ~ s(year, 4) + s(age, 5) + education)` (backfitting).

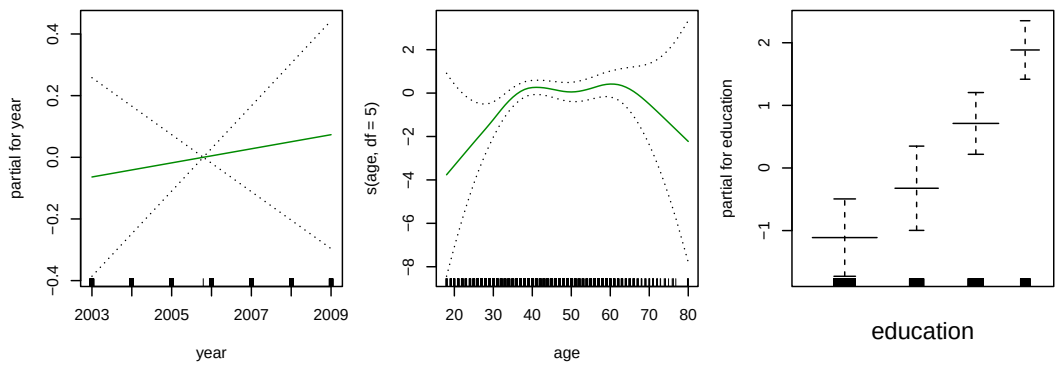
Pros. Automatic nonlinear f_j, often better prediction, per-variable effects, smoothness summarised by df. **Cons.** Additivity misses interactions; add $X_j \times X_k$ or $f_{jk}(X_j, X_k)$ by hand, or use random forests and boosting.

GAMs for classification. For binary Y with $p(X) = P(Y = 1 \mid X)$, the logistic GAM is

$$\log \frac{p(X)}{1 - p(X)} = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p).$$

For wage above \$250 000:

```
gam_lr <- gam(I(wage > 250) ~ year + s(age, df = 5) + education,
              family = binomial, data = Wage, subset = (education != "1. < HS Grad"))
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot(gam_lr, se = TRUE, col = "green4")
```



The lowest education level has no high earners and is excluded (separation, Lecture 9) [Supplement].

Summary. Regression splines: basis functions; smoothing splines: loss plus penalty; GAMs: $x_j \to f_j(x_j)$, additively.

10.7 Selected exercises

Motivation. Exact degrees of freedom of the lecture's smoothers.

Exercise 10.1 (dimension of spline spaces)

Let $\xi_1 < \cdots < \xi_K$ be knots.
(a) Show that the functions $1, x, x^2, x^3, (x-\xi_1)_+^3, \ldots, (x-\xi_K)_+^3$ are linearly independent, so that cubic splines with these knots form a space of dimension $K+4$.
(b) A *natural* cubic spline is in addition linear on $(-\infty, \xi_1]$ and on $[\xi_K, \infty)$. For $K \geq 2$, show that natural cubic splines form a space of dimension K. (This is why `ns(age, df = 4)`, which uses three interior and two boundary knots, has four columns besides the intercept.)

Solution [Supplement].

(a) Suppose $f(x) = \sum_{j=0}^{3} c_j x^j + \sum_{k=1}^{K} \theta_k (x-\xi_k)_+^3 = 0$ for all x. On $(-\infty, \xi_1)$ all truncated terms vanish, so a cubic polynomial is zero on an interval and $c_0 = \cdots = c_3 = 0$. Inductively, if $\theta_1 = \cdots = \theta_{k-1} = 0$, then on (ξ_k, ξ_{k+1}) (with $\xi_{K+1} = \infty$) $f(x) = \theta_k (x-\xi_k)^3 = 0$, so $\theta_k = 0$. The notes showed that these functions span the cubic splines, so they form a basis.

(b) Write a cubic spline in the basis of (a).

Step 1. On $(-\infty, \xi_1)$, $f = c_0 + c_1 x + c_2 x^2 + c_3 x^3$, which is linear iff $c_2 = c_3 = 0$.

Step 2. On (ξ_K, ∞) every truncated term is active, and the coefficients of x^3 and x^2 are $c_3 + \sum_k \theta_k$ and $c_2 - 3\sum_k \theta_k \xi_k$. Given $c_2 = c_3 = 0$, linearity there means $\sum_k \theta_k = 0$ and $\sum_k \theta_k \xi_k = 0$.

Step 3. The four constraints are linear. The first two involve only c_2, c_3; the last two involve only θ, with coefficient vectors $(1, \ldots, 1)$ and $(\xi_1, \ldots, \xi_K)$, which are linearly independent because the knots are distinct and $K \geq 2$. So the constraints are independent and the dimension is $(K+4) - 4 = K$. ∎

Exercise 10.2 (degrees of freedom of a step function)

Let every bin of the step-function regression of Section 10.2 contain at least one observation, and let n_j be the size of bin $j = 0, \ldots, K$. Show that the hat matrix H (with $\hat{y} = Hy$) replaces each y_i by the mean of its bin, and that $\operatorname{tr} H = K + 1$, the number of coefficients. Why must C_0 be left out when an intercept is included?

Solution [Supplement].

Step 1. Since $C_0 + \cdots + C_K \equiv 1$, the columns $1, C_0, \ldots, C_K$ are linearly dependent and least squares would not have unique coefficients; dropping C_0 leaves $K+1$ independent columns spanning the same space as $C_0, \ldots, C_K$.

Step 2. The fitted function is constant on each bin, with value $a_0 = \beta_0$ on bin 0 and $a_j = \beta_0 + \beta_j$ on bin $j \geq 1$ (a bijection between the a's and the β's), and $\mathrm{RSS} = \sum_j \sum_{i \in \text{bin } j} (y_i - a_j)^2$ separates over bins. By the constant-model result of Lecture 5 each a_j is minimised by the bin mean $\bar{y}_{(j)}$.

Step 3. Ordering the observations by bin, H is block diagonal with blocks $n_j^{-1} \mathbf{1}\mathbf{1}^{\mathsf{T}}$, the matrix that averages within the bin. Each block has trace $n_j \cdot n_j^{-1} = 1$, so $\operatorname{tr} H = K + 1$. This agrees with the definition $\mathrm{df} = \operatorname{tr} S$ used for smoothing splines. ∎

Exercise 10.3 (effective degrees of freedom decrease with λ)

With $n \geq 3$ distinct values x_i, the smoothing-spline fitted values can be written $\hat{g}_\lambda = S_\lambda y$ with $S_\lambda = (I + \lambda Q)^{-1}$, where Q is a fixed symmetric positive semidefinite $n \times n$ matrix whose null space consists of the vectors $(a + bx_i)_{i=1}^n$ (a standard fact from the theory of smoothing splines; take it as given). Show that

$$\mathrm{df}_\lambda = \mathrm{tr} S_\lambda = \sum_{k=1}^{n} \frac{1}{1 + \lambda d_k},$$

where $d_k \geq 0$ are the eigenvalues of Q, that df_λ is strictly decreasing in λ, and that it tends to n as $\lambda \to 0$ and to 2 as $\lambda \to \infty$.

Solution [Supplement].

Step 1. Write $Q = U \,\mathrm{diag}(d_1, \ldots, d_n) U^\mathsf{T}$ with U orthogonal. Then $I + \lambda Q = U \,\mathrm{diag}(1 + \lambda d_k) U^\mathsf{T}$ has positive eigenvalues, and $S_\lambda = U \,\mathrm{diag}\{1/(1 + \lambda d_k)\} U^\mathsf{T}$. The trace is invariant under orthogonal similarity, so $\mathrm{df}_\lambda = \sum_k 1/(1 + \lambda d_k)$.

Step 2. The null space of Q is spanned by $(1, \ldots, 1)$ and $(x_1, \ldots, x_n)$, which are independent since the x_i are not all equal. So exactly two eigenvalues are 0 and $n - 2 \geq 1$ are positive.

Step 3. $\frac{d}{d\lambda} \mathrm{df}_\lambda = -\sum_k d_k/(1 + \lambda d_k)^2 < 0$, since some $d_k > 0$. As $\lambda \to 0$ every term tends to 1; as $\lambda \to \infty$ the terms with $d_k > 0$ tend to 0 and the two with $d_k = 0$ stay 1.

Step 4. The eigenvalues $1/(1 + \lambda d_k) \in (0, 1]$: linear trends ($d_k = 0$) are kept, rough components (large d_k) shrunk most, as in ridge. ■

Lecture 11. Causal Inference

11.1 Why causation

Motivation. Causal questions ask what Y would be if we changed something; answering needs assumptions on treatment assignment.

Causal questions: is smoking harmful? Would cleaner air reduce respiratory deaths? A **confounder** X affects both treatment Z and outcome Y (hot weather drives ice-cream sales and drownings).

Job training (LaLonde). Did a 1976 programme raise 1978 earnings? The groups differ before treatment:

```
lalonde %>%
  group_by(treat) %>%
  summarise(across(c(age, educ, married, re74, re75, re78), mean), black = mean(race ==
    "black"))
```

```
## # A tibble: 2 x 8
##   treat   age  educ married  re74  re75  re78 black
##   <int> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     0  28.0  10.2   0.513 5619. 2466. 6984. 0.203
## 2     1  25.8  10.3   0.189 2096. 1532. 6349. 0.843
```

The treated are more often Black, less often married, and earned far less in 1974–75.

Smoking and lung cancer. Lung cancer tripled by the mid-1940s; observational studies followed in 1950 (Wynder and Graham; Doll and Hill; Hammond). Hammond matched 36 975 heavy smokers to non-smokers on many covariates. In 122 discordant pairs the smoker died in 110; with no effect $X \sim B(122, 1/2)$:

```
binom.test(110, 122, p = 0.5, alternative = "greater")$p.value
```

```
## [1] 2.733658e-21
```

The weak point is an *unmeasured* confounder (Fisher: a gene causing both).

Counterfactuals. Zeus dies five days after a heart transplant; had he lived without it, the transplant caused his death. Causal effects compare what happened with what *would have happened*.

11.2 Potential outcomes

Motivation. Potential outcomes give notation in which both treated and untreated outcomes exist.

For units $i = 1, \dots, N$: covariates X_i; treatment $Z_i \in \{0, 1\}$; **potential outcomes** $Y_i(1)$ (if treated) and $Y_i(0)$ (if not); and the observed outcome

$$Y_i = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

For Zeus, $Z_i = 1$ and $Y_i = 1$ (death), with $Y_i(1) = 1$, $Y_i(0) = 0$: $Y_i(1)$ is *factual*, $Y_i(0)$ *counterfactual*. (Also written treatment A and potential outcomes $Y^{a=1}$, $Y^{a=0}$.)

Hidden assumptions:

- **no interference**: Y_i depends only on Z_i;
- **consistency**: $Y_i = Y_i(1)$ if $Z_i = 1$ (**no multiple versions of treatment**).

Together these are the **stable unit treatment value assumption** (SUTVA).

Causal effects. The individual effect is $\tau_i = Y_i(1) - Y_i(0)$; the **average treatment effect** is

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

Other measures: the risk ratio $\mathbb{E}[Y_i(1)]/\mathbb{E}[Y_i(0)]$ and, for binary outcomes, the odds ratio $\frac{P(Y_i(1)=1)\,P(Y_i(0)=0}{P(Y_i(1)=0)\,P(Y_i(0)=1}$

The **sample** ATE is $\frac{1}{N}\sum_i\{Y_i(1) - Y_i(0)\}$, the **population** ATE is $\mathbb{E}[Y_i(1) - Y_i(0)]$, and SATE $\to$ PATE as $N \to \infty$.

Science table. Both potential outcomes per unit; the data show one, the other is "?".

The fundamental problem of causal inference (Holland, 1986): τ_i is never observed. The ATE can be identified under random assignment or unconfoundedness, with **overlap** $0 < P(Z = 1 \mid X) < 1$.

11.3 Randomised experiments and observational studies

Motivation. Randomisation balances everything; without it, comparable groups must be built from covariates.

Randomised experiments (trials, A/B tests) are the gold standard but often infeasible; observational data are easy to get, hard to analyse. Randomisation balances observed and unobserved covariates and is the "reasoned basis for inference" (Fisher, 1935):

$$(Y(1), Y(0)) \perp\!\!\!\perp Z.$$

Then $\mathbb{E}[Y(1)] = \mathbb{E}[Y(1) \mid Z = 1] = \mathbb{E}[Y(1) \mid Z = 0]$, and $Y(1)$ is observed when $Z = 1$, so

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0] = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] = \bar{Y}_t - \bar{Y}_c,$$

with no distributional assumption (the **identification assumption** is the independence).

When it fails. If a doctor treats patients with high expected $Y_i(0)$, then $\mathbb{E}[Y(0) \mid Z = 1] > \mathbb{E}[Y(0) \mid Z = 0]$: the medical history X confounds. **Overt bias** (observed X) is removed by regression, matching, weighting; **hidden bias** needs special designs (instruments, discontinuities, difference-in-differences) and sensitivity analysis.

11.3.1 Unconfoundedness

The key assumption is

$$(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X \qquad \text{(unconfoundedness, ignorability)}:$$

treatment is random among units with the same X. Then

$$\mathbb{E}[Y(1)] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X]\big] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X, Z = 1]\big] = \mathbb{E}\big[\mathbb{E}[Y \mid X, Z = 1]\big],$$

and likewise for $Y(0)$. With outcome models $\hat{\mu}_1$ (treated) and $\hat{\mu}_0$ (controls),

$$\hat{\tau}^{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n}\{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}.$$

In general $\mathbb{E}[Y(1)] \neq \mathbb{E}[Y \mid Z = 1]$.

Connection to regression. The classical adjustment $\mathbb{E}(Y \mid X) = \beta_0 + \beta_1 Z + \beta_2 X$ corresponds to $\mathbb{E}[Y(1) \mid X] = \beta_0 + \beta_1 + \beta_2 X$ and $\mathbb{E}[Y(0) \mid X] = \beta_0 + \beta_2 X$, so the ATE is β_1. But the two means may need different forms, $\tau(X)$ may vary with X, poor overlap means *extrapolation*, and adjusting while seeing outcomes invites cherry-picking.

11.4 Propensity score methods

Motivation. The propensity score compresses many covariates into one number that suffices for comparison.

The **propensity score** $e(X_i) = P(Z_i = 1 \mid X_i)$ (Rosenbaum and Rubin, 1983):

Theorem 11.1 (balancing property)

If $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X$, then $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid e(X)$.

Proof [Supplement]. It suffices to show $P(Z = 1 \mid Y(1), Y(0), e(X)) = e(X)$, which does not depend on the potential outcomes. By the tower property, $P(Z = 1 \mid Y(1), Y(0), e(X)) = \mathbb{E}[P(Z = 1 \mid Y(1), Y(0), X) \mid Y(1), Y(0), e(X)] = \mathbb{E}[e(X) \mid Y(1), Y(0), e(X)] = e(X)$, using unconfoundedness in the middle step and that $e(X)$ is a function of X. ■

Sharing $e(X)$ is enough. In observational studies it is estimated, usually by logistic regression, $\operatorname{logit} P(Z_i = 1 \mid X_i) = \beta_X^\mathsf{T} X_i$ and $\hat{e}(X_i) = \operatorname{expit}(\hat{\beta}_X^\mathsf{T} X_i)$. then used to stratify, match or weight.

11.4.1 Stratification

(1) Estimate $\hat{e}(X_i)$; (2) cut it into K narrow strata (quantiles or $(0, \frac{1}{K}], \ldots, (\frac{K-1}{K}, 1)$); (3) treat each stratum as a randomised experiment:

$$\hat{\tau} = \sum_{k=1}^{K} (\bar{Y}_{k1} - \bar{Y}_{k0}) \, \frac{n_k}{n},$$

with $\bar{Y}_{k1}$, $\bar{Y}_{k0}$ the treated and control means in stratum k of size n_k. Equivalently, a treated unit in stratum k has weight $\frac{1}{n_{k1}} \frac{n_k}{n}$ and a control $\frac{1}{n_{k0}} \frac{n_k}{n}$, and a weighted regression $Y = \alpha + \beta Z + \epsilon$ gives $\hat{\tau} = \hat{\beta}$.

11.4.2 Matching

1:1 matching (fewer treated than controls): (1) match each treated unit to a control with similar X or $e(X)$, with or without replacement (bias–variance); (2) check balance; (3) discard unmatched controls; (4) analyse as a paired experiment: mean treated-minus-control difference (or regress Y on Z), randomisation test of no effect.

11.4.3 Inverse probability weighting

IPW weights treated units by $1/e(X)$, controls by $1/(1-e(X))$: a treated person with $e(X) = 1/5$ stands for five "clones". The **pseudo-population** has $(Y(1), Y(0)) \perp\!\!\!\perp Z$.

Proposition 11.2 (IPW identification)

Under unconfoundedness and overlap,

$$\tau = \mathbb{E}\Big[\frac{Z_i Y_i}{e(X_i)}\Big] - \mathbb{E}\Big[\frac{(1 - Z_i) Y_i}{1 - e(X_i)}\Big].$$

Proof. First term: $\mathbb{E}[\frac{Z_i Y_i}{e(X_i)}] = \mathbb{E}[\frac{\mathbb{E}[Z_i Y_i \mid X_i]}{e(X_i)}] = \mathbb{E}[\frac{e(X_i)\,\mathbb{E}[Y_i \mid Z_i = 1, X_i]}{e(X_i)}] = \mathbb{E}[\mathbb{E}[Y_i(1) \mid Z_i = 1, X_i]] = \mathbb{E}[\mathbb{E}[Y_i(1) \mid X_i]] = \mathbb{E}[Y_i(1)]$, using $\mathbb{E}[Z_i Y_i \mid X_i] = P(Z_i = 1 \mid X_i)\mathbb{E}[Y_i \mid Z_i = 1, X_i]$, consistency and unconfoundedness. The second term equals $\mathbb{E}[Y_i(0)]$ in the same way [Supplement]. ■

With estimated scores $\hat{e}_i$,

$$\hat{\tau}^{\text{ATE}} = \frac{1}{n}\sum_i \frac{Z_i Y_i}{\hat{e}_i} - \frac{1}{n}\sum_i \frac{(1-Z_i)Y_i}{1-\hat{e}_i},$$

or a regression of Y on Z with weights $w_i = Z_i/\hat{e}_i + (1-Z_i)/(1-\hat{e}_i)$, unstable for $\hat{e}_i$ near 0 or 1. ATT weights: 1 (treated), $\hat{e}_i/(1-\hat{e}_i)$ (controls; Exercise 11.3).

11.5 Example: job training and earnings

Packages: `MatchIt`, `WeightIt`; balance by `cobalt::bal.tab()`, effects by `marginaleffects::avg_comparisons()`. Data: `cobalt::lalonde` (185 treated, 429 controls).

```
no.adj <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = NULL, distance = "glm")
summary(no.adj)$sum.all[, 1:3]
```

```
##            Means Treated Means Control Std. Mean Diff.
## distance    5.774355e-01     0.1822248      1.79408605
## age         2.581622e+01    28.0303030     -0.30944526
## educ        1.034595e+01    10.2354312      0.05496466
## raceblack   8.432432e-01     0.2027972      1.76154189
## racehispan  5.945946e-02     0.1421911     -0.34984254
## racewhite   9.729730e-02     0.6550117     -1.88186760
## married     1.891892e-01     0.5128205     -0.82630927
## nodegree    7.081081e-01     0.5967366      0.24497023
## re74        2.095574e+03  5619.2365064     -0.72108381
## re75        1.532055e+03  2466.4844431     -0.29026291
```

The standardised mean differences are huge (1.76 Black, −1.88 White, −0.72 1974 earnings); a rule of thumb is $|\text{SMD}| < 0.1$ [Supplement]. Three adjustments:

```
ps.str <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = "subclass", distance = "glm", link = "probit"
    , subclass = 7)
ps.mat <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = "nearest", distance = "glm", link = "probit",
     replace = TRUE)
W <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
              data = lalonde, method = "glm", estimand = "ATT")
round(cbind(stratified = summary(ps.str, un = FALSE)$sum.across[, 3],
            matched = summary(ps.mat, un = FALSE)$sum.matched[, 3]), 3)
```

```
##            stratified matched
## distance        0.063   0.003
## age            -0.036  -0.085
## educ            0.027  -0.083
## raceblack       0.084   0.015
## racehispan     -0.018   0.046
## racewhite      -0.089  -0.055
## married        -0.030   0.097
## nodegree        0.027   0.143
## re74           -0.089  -0.013
## re75           -0.074  -0.017
```

```
bal.tab(W, un = TRUE)
```

```
## Balance Measures
##                 Type Diff.Un Diff.Adj
## prop.score  Distance  1.7941  -0.0205
## age          Contin. -0.3094   0.1188
## educ         Contin.  0.0550  -0.0284
## race_black    Binary  0.6404  -0.0022
## race_hispan   Binary -0.0827   0.0002
```

```
## race_white    Binary -0.5577   0.0021
## married       Binary -0.3236   0.0186
## nodegree      Binary  0.1114   0.0184
## re74         Contin. -0.7211  -0.0021
## re75         Contin. -0.2903   0.0110
##
## Effective sample sizes
##            Control Treated
## Unadjusted  429.       185
## Adjusted     99.82     185
```

All three nearly balance (7 strata; matching keeps 84 distinct controls; IPW keeps everyone). ATT by weighted regression of 1978 earnings on treatment:

```
str.data <- match.data(ps.str); mat.data <- match.data(ps.mat)
est <- function(fit, ...) {
  a <- avg_comparisons(fit, variables = "treat", ...)
  round(c(estimate = a$estimate, low = a$conf.low, high = a$conf.high))
}
fit.str <- lm(re78 ~ treat, data = str.data, weights = weights)
fit.mat <- lm(re78 ~ treat, data = mat.data, weights = weights)
lalonde$ipw.weight <- W$weights
fit.ipw <- lm(re78 ~ treat, data = lalonde, weights = ipw.weight)
rbind(
  stratification = est(fit.str, vcov = ~subclass, newdata = subset(str.data, treat == 1)
    , wts = "weights"),
  matching       = est(fit.mat, vcov = "HC3", newdata = subset(mat.data, treat == 1),
    wts = "weights"),
  ipw            = est(fit.ipw, vcov = "HC3", wts = "ipw.weight"),
  naive          = round(c(coef(lm(re78 ~ treat, data = lalonde))[2], NA, NA)))
```

```
##                estimate  low high
## stratification     1301 -423 3025
## matching           1290 -603 3184
## ipw                1214 -416 2844
## naive              -635   NA   NA
```

Full matching (no units discarded):

```
ps.full <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                   data = lalonde, method = "full", distance = "glm", link = "probit")
full.data <- match.data(ps.full)
fit.full <- lm(re78 ~ treat, data = full.data, weights = weights)
est(fit.full, vcov = ~subclass, newdata = subset(full.data, treat == 1), wts = "weights"
    )
```

```
## estimate      low     high
##     1855      364     3346
```

Naive: the programme *lowered* earnings; adjusted: about \$1 200–1 900. The original experiment found \$2 114, 95% CI (458, 3955), closest to full matching.

Take-home. Matching and weighting separate *design* (no outcomes) from *analysis* (compare means); try several methods and check agreement.

11.6 Selected exercises

Motivation. Why the naive comparison had the wrong sign and why the adjustments agree.

Throughout, consistency and SUTVA hold, $e(X) = P(Z = 1 \mid X)$, and expectations are finite.

Exercise 11.1 (the naive comparison)

Show that

$$\mathbb{E}[Y \mid Z=1]-\mathbb{E}[Y \mid Z=0]=\underbrace{\mathbb{E}[Y(1)-Y(0) \mid Z=1]}_{\text{ATT}}+\underbrace{\mathbb{E}[Y(0) \mid Z=1]-\mathbb{E}[Y(0) \mid Z=0]}_{\text{selection bias}},$$

that the bias vanishes and ATT = ATE under random assignment, and interpret the sign of the bias in the job-training data.

Solution [Supplement].

Step 1. By consistency, $\mathbb{E}[Y \mid Z=1]=\mathbb{E}[Y(1) \mid Z=1]$ and $\mathbb{E}[Y \mid Z=0]=\mathbb{E}[Y(0) \mid Z=0]$. Adding and subtracting $\mathbb{E}[Y(0) \mid Z=1]$ gives the decomposition.

Step 2. If $(Y(1), Y(0)) \perp\!\!\!\perp Z$, then $\mathbb{E}[Y(0) \mid Z=1]=\mathbb{E}[Y(0) \mid Z=0]=\mathbb{E}Y(0)$, so the bias is 0, and likewise $\mathbb{E}[Y(1)-Y(0) \mid Z=1]=\mathbb{E}[Y(1)-Y(0)]$.

Step 3. Naive difference negative, adjusted ATT +\$1 200 to +\$1 900: the selection bias is strongly negative (participants would have earned much less anyway, as 1974–75 earnings show). ■

Exercise 11.2 (stratification is weighting)

Divide n units into strata $k=1, \ldots, K$ of sizes n_k, with $n_{k1} \geq 1$ treated and $n_{k0} \geq 1$ control units, and estimate the propensity of every unit in stratum k by $\hat{e}_k=n_{k1}/n_k$. Show that the IPW estimator

$$\hat{\tau}^{\mathrm{IPW}}=\frac{1}{n} \sum_i \frac{Z_i Y_i}{\hat{e}_i}-\frac{1}{n} \sum_i \frac{(1-Z_i) Y_i}{1-\hat{e}_i}$$

equals the stratified estimator $\sum_k(\bar{Y}_{k1}-\bar{Y}_{k0})\, n_k/n$ of Section 11.4.

Solution [Supplement]. Group the sums by stratum. For the treated part,

$$\frac{1}{n} \sum_k \frac{n_k}{n_{k1}} \sum_{i \in k,\, Z_i=1} Y_i=\frac{1}{n} \sum_k n_k \bar{Y}_{k1}=\sum_k \frac{n_k}{n} \bar{Y}_{k1}.$$

For the control part, $1-\hat{e}_k=n_{k0}/n_k$, and the same computation gives $\sum_k(n_k/n)\bar{Y}_{k0}$. Subtracting yields the stratified estimator: IPW with propensity estimated by the stratum's treated share. ■

Exercise 11.3 (weights for the effect on the treated)

Assume $Y(0) \perp\!\!\!\perp Z \mid X$, $e(X)<1$ and $P(Z=1)>0$. Prove that

$$\mathbb{E}[Y(0) \mid Z=1]=\frac{\mathbb{E}\big[(1-Z) \frac{e(X)}{1-e(X)} Y\big]}{P(Z=1)},$$

and deduce that the ATT is the difference between the mean outcome of the treated (weight 1) and the weighted mean outcome of the controls with weights $e(X)/(1-e(X))$.

Solution [Supplement].

Step 1. Condition on X. By consistency $(1-Z)Y=(1-Z)Y(0)$, and by unconfoundedness

$$\mathbb{E}\Big[(1-Z) \frac{e(X)}{1-e(X)} Y \,\Big|\, X\Big]=\frac{e(X)}{1-e(X)}\, \mathbb{E}[1-Z \mid X]\, \mathbb{E}[Y(0) \mid X]=e(X)\, \mathbb{E}[Y(0) \mid X].$$

Step 2. Again by unconfoundedness, $e(X)\mathbb{E}[Y(0) \mid X]=\mathbb{E}[Z \mid X]\, \mathbb{E}[Y(0) \mid X]=\mathbb{E}[ZY(0) \mid X]$. Taking expectations, the numerator equals $\mathbb{E}[ZY(0)]=P(Z=1)\, \mathbb{E}[Y(0) \mid Z=1]$, which is the claim.

Step 3. With $w = 1$ for treated units and $w = e(X)/(1 - e(X))$ for controls, the same argument with Y replaced by 1 gives $\mathbb{E}[(1-Z)w] = P(Z = 1)$, so $\mathbb{E}[Y(0) \mid Z = 1] = \mathbb{E}[(1-Z)wY]/\mathbb{E}[(1-Z)w]$, a weighted control mean. Since $\mathbb{E}[Y(1) \mid Z = 1] = \mathbb{E}[Y \mid Z = 1]$, the ATT is the treated mean minus this weighted control mean. ■